Tectonophysics - Elsevier Publishing Company, Amsterdam
Printed in The Netherlands

PALEOMAGNETIC EVIDENCE FOR THE ROTATION OF THE IBERIAN PENINSULA[1]

R. VAN DER VOO

Palaeomagnetic Laboratory, State University Utrecht, Utrecht (The Netherlands)

(Received July 2, 1968)
(Resubmitted November 29, 1968)

SUMMARY

The results of a paleomagnetic investigation on igneous and sedimentary rocks from Portugal and Spain are presented. The age of the formations investigated varies from Ordovician to Eocene. Apart from geologic studies the Natural Remanent Magnetizations (N.R.M.) were analyzed with the aid of a.c. magnetic field and thermal demagnetization techniques. In the case of three folded formations the characteristic remanent magnetizations could be proved to be acquired before the subsequent folding took place. These formations are the Upper Silurian Almadén volcanics, the Upper Carboniferous–Lower Permian Buçaco Formation of Portugal and the Eocene basalts of the Lisbon region. All three other Upper Carboniferous–Lower Permian sample groups of Spain yield similar directions of magnetization. Several groups of Paleozoic and Triassic samples revealed only secondary magnetizations.

Together with previous results from the Spanish Meseta and the Spanish Pyrenees, the data are compared with results from Africa and from other European countries. The comparison is satisfactory only for the Upper Carboniferous–Lower Permian results: it indicates that the Iberian Peninsula has rotated relative to that part of Europe north of the Alpine fold belts. This rotation has been counterclockwise over approximately 35°. It is argued that a plausible ancient configuration can be realised by rotating the Iberian Peninsula back to its Permian position, while closing the Bay of Biscay at the 2,000-m depth line. The pivot point of this rotation lies in the western Pyrenees, as previously suggested by Carey (1958).

INTRODUCTION

General outline

Since more and more valuable data have become available in different domains of the earth sciences, such as oceanography, seismology, heat-flow measurements and paleomagnetism, a revival of many geotectonic theories

[1]This contribution has been presented as a doctoral thesis in the Faculty of Mathematics and Sciences, State University, Utrecht.

can be recognized in the last decades. Especially, mutual displacements of the continents, or parts of the continents, were indicated by paleomagnetism. Postulating the existence of a dipolar geocentric axial geomagnetic field in geologic history, paleomagnetic data are known to supply information on distance (latitude) and orientation of a continental block with respect to an ancient pole.

Even without being able to measure the ancient longitudinal positions of the continents, one can in this way often check many of the continental-drift theories, when sufficient paleomagnetic data become available.

In this view, the complicated Mediterranean situation offers an outstanding and very promising object for paleomagnetic research. The geotectonic relationships of the Alpine orogenic belts and the various stable blocks have recently enjoyed increasing interest.

In this program the State University of Utrecht has initiated various studies. Of these I mention the Estérel region (Zijderveld, in preparation (a)), the Southern Alpine realm (Dietzel, 1960; Van Hilten, 1960; De Boer, 1963; Van Hilten and Zijderveld, 1966; Zijderveld and De Jong, 1969), Sardinia (De Jong and Zijderveld, 1969; Zijderveld et al., in preparation), Turkey (Gregor and Zijderveld, 1964; Van der Voo, 1968a) and Lebanon (Van Dongen et al., 1967). A very interesting subject, moreover, was supposed to be the paleomagnetism of the Spanish Meseta, and its related Alpine belts, viz., the Betic Cordillera, the Catalanides and the Pyrenees. Many theoreticians have already reported their views based on geologic data alone. Carey (1958) suggested a counterclockwise rotation of 30 or 40° of the Iberian block around a pivot point in the western Pyrenees, together with an opening of the so-called Biscay Sphenochasm. Bullard et al. (1965) formulated a similar hypothesis in their reconstruction by closure of the Atlantic Ocean. Moreover, Mattauer (1968) published some ideas on observed right-lateral displacements in the Pyrenees, whereas Carey (1958) takes the movements to be left-lateral.

Previous paleomagnetic investigations in Spain

Clegg et al. (1957) have started paleomagnetic work on the Spanish Meseta. They reported magnetic directions approximately parallel to the recent local geomagnetic field, in Triassic redbeds from northwest and central Spain. Eight years ago two research students, completing a Utrecht doctoral thesis in the central Pyrenees on the geology of Upper Palaeozoic formations, mentioned Permo-Triassic paleomagnetic directions (Van der Lingen, 1960; Schwarz, 1962,1963). Recently Van Dongen (1967) published his results from Lower Permian andesites and Triassic redbeds of the eastern Pyrenees. All three of these investigators found virtual pole positions systematically diverging from contemporaneous ones, found for that part of Europe which has remained stable since the Uppermost Permian, in this study to be called "stable" Europe.

While the present study was in preparation, a paper was published by Watkins and Richardson (1968), who argued that evidence from the Lisbon basalt flows pointed to post-Eocene movements of the Iberian Peninsula relative to stable Europe. However, as Van der Voo (1968c) pointed out, it is not likely that Watkins and Richardson's mean direction represents the true Eocene geomagnetic field direction in Portugal.

History and purposes of the study

The present study has been stimulated by the theories mentioned above and is intended as a contribution to the discussion in providing paleomagnetic data from the Spanish Meseta for various geologic periods. It was started in 1962 and several sampling trips were made in the last six years, under supervision of Professors M.G. Rutten and J. Veldkamp.

The Triassic redbeds, occurring on the margins of the Spanish Meseta, initially looked very promising. After it became clear, however, that these rocks failed to provide original directions of magnetization (Van der Voo, 1968b), special attention was paid to Paleozoic rocks. In Portugal, finally, some Upper Mesozoic–Lower Tertiary volcanic rocks have been collected in order to obtain some information on the time of the rotation.

Table I lists all regions, rock types and ages for the formations from which samples were collected. In the following chapters the results of these and previous studies will be discussed, and they will be compared with coeval data from stable Europe and Africa. In the last chapter, finally, an outline is given of the conclusions that can be drawn.

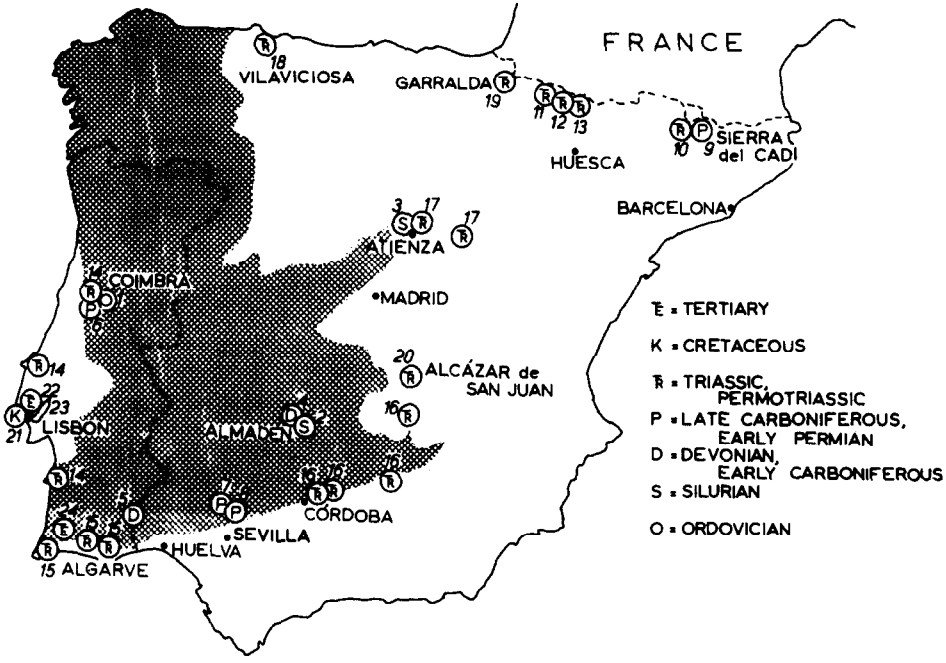
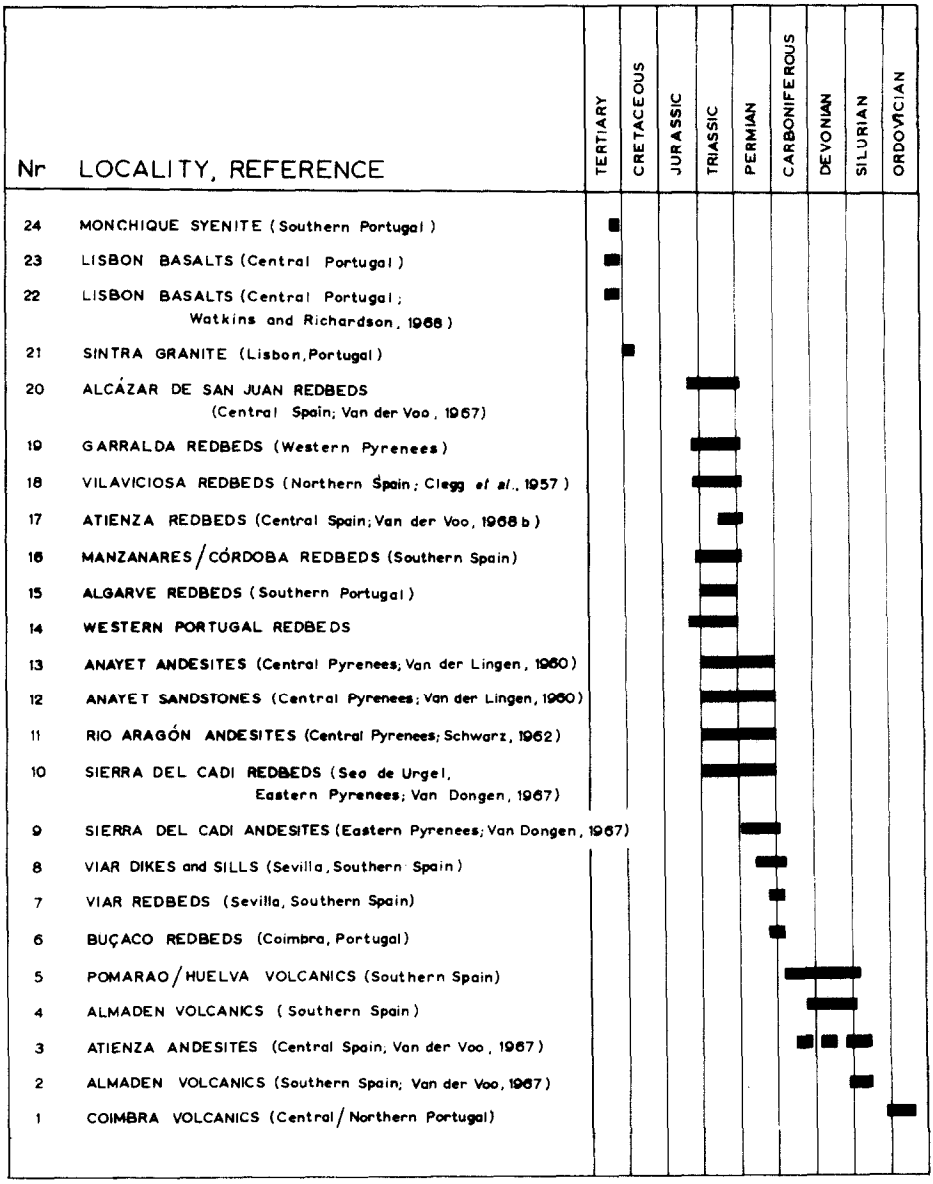


Fig.1. Map of the Iberian Peninsula with the sampling areas indicated. Numbers refer to Table I. The Paleozoic of the Iberian Meseta has been cross-hatched.

TABLE I

Sampling regions, formations and ages of the Iberian paleomagnetic studies described in this paper



METHODS OF RESEARCH

Sampling

The samples (Table I) were collected in the time-span 1962–1967. Their orientations were determined with the aid of a Caminada and Tamson clinometer compass. The corrections for the geomagnetic variation, varying between 5° and 10° west, have been applied afterwards. In most of the sampling areas the sites were widely separated. They were deliberately chosen from different beds and flows in order to eliminate as far as possible the influence of secular variation.

Measurements and demagnetization

The samples were sawn to approximately equidimensional shape and embedded in their correct orientation in cubes of paraffin with 10-cm edges. Thereupon, directions and intensities were measured on the astatic magnetometers of the Paleomagnetic Laboratory in Utrecht. The natural remanent magnetization (N.R.M.) of all samples was further analyzed by progressive demagnetization with a.c. magnetic fields up to 3,000 Oe (peak value). For an extensive description of the methods of measurement and the analysis of the N.R.M. with the aid of a.c. magnetic field demagnetization the reader is referred to As and Zijderveld (1958), As (1960), Van Everdingen (1960) and Zijderveld (1967a).

Furthermore, several samples of each group were subjected to thermal demagnetization. For this treatment cores were drilled (diameter 25 mm, height 22 mm) from the hand samples oriented in the paraffin cubes. The demagnetization was carried out stepwise, the samples being heated and cooled again to room temperature in a furnace placed in a zero ambient field. The directions and intensities of the samples, being remeasured after each step, yielded extra information besides the a.c. magnetic field demagnetization, for instance on blocking temperatures and constitution of the N.R.M. For an extensive description of the furnace, I refer to Mulder and Zijderveld (in preparation).

The numerical data, obtained in a.c. magnetic field and thermal demagnetization, were further analyzed with the aid of various programmed operations by the x.8 computer of the Mathematical Department in Utrecht (Klootwijk, 1967). They are summarized in Table II. In the next section equal area projections of the directions of magnetization are presented. Anyone wishing to receive the numerical data per sample may obtain a set of tables directly from the author.

LOCALITIES AND ANALYSIS OF THE N.R.M.

In this section all collected samples will be dealt with in chronological order. They are listed in Table I. Fig.1 is a regional map of the Iberian Peninsula with the sampling localities indicated.

Special interest has been given to the formation of Permian and Triassic age, since from stable (extra-Alpine) Europe, as well as from stable Africa,

TABLE II

Mean directions of magnetization and pole positions of the Iberian rocks

No.[1]	Formation	Age[2]	D(°)	I(°)	α_{95}(samp.)[3]	α_{95}(loc.)	Pole position
24	Monchique Syenite	Te, 57 m.y.	182	-37	6.5 (8)	14 (2)	73°N 165.5°E
23	Lisbon Basalts	Te	354	+40.5	4.5 (19)	8.5 (5)	73.5°N 170°W
21	Sintra Granite	Ku, 80 m.y.	359	+43.5	5 (25)	8 (8)	76.5°N 174°E
20	Alcázar de San Juan redbeds	Tr	359.5	+23	6 (39)	- (2)	63°N 177.5°E
19	Garralda redbeds	Tr	350	+18	density distribution		
13	Anayet Andesites, redbeds	P-Tr	164	-14	10 (11)	- -	52°N 154°W
11	Rio Aragón Andesites	P-Tr	152	-22.5	6 (14)	- -	51°N 133°W
10	Sierra del Cadi redbeds	Tr	340.5	+24	11 (4)	- (1)	54.5°N 142°W
9	Sierra del Cadi Andesites	P (Pl ?)	169.5	- 3	4 (41)	6 (10)	48.5°N 163°W
8	Viar dikes and sills	Cu-Pl	155.5	+10.5	4 (14)	13 (3)	41°N 152°W
7	Viar redbeds	Cu-Pl	151	+ 2	4.5 (8)	6 (3)	42.5°N 144°W
6	Buçaco redbeds	Cu-Pl	149	+11	5 (17)	7 (4)	35.5°N 148.5°W
3	Atienza Andesites	Su (D-C ?)	159	+18.5	5 (33)	12 (6)	35.5°N 157°W
2	Almadén volcanics	Su	130.5	+22.5	11 (10)	- (2)	21°N 132°W
1	Coimbra volcanics	Ou	101	+16.5	- (2)	- (1)	-

[1]The numbers correspond with Table I and Fig.1.

[2]Te = Eocene; K = Cretaceous; Tr = Triassic; P = Permian; C = Carboniferous; D = Devonian; S = Silurian; O = Ordovician; u = upper; l = lower.

[3]α_{95} given for localities (loc.) and samples (samp.) used in the analysis.

reliable data are available for comparison. The older Paleozoic rocks have undergone both Hercynian and Alpine orogenies. They were preliminarily investigated to find out whether some theories might be extended to earlier geologic times. The Late Mesozoic and Early Tertiary periods might give evidence for the time-span of the postulated geotectonic movements.

Ordovician and Silurian

Previous investigations

In an earlier presentation of part of this study (Van der Voo, 1967) I described the paleomagnetism of two sample groups of volcanics from Almadén and Atienza (Table I, no.2,3). The directions of magnetization of the well-dated Upper Silurian Almadén volcanics (10 samples, 2 sites, 2 flows) have been analyzed with the aid of a.c. magnetic field demagnetizations. One of the two components, which were present in all samples, had a direction parallel to the present-day geomagnetic field and is likely to be secondary. The difference in attitude of the two flows permits the application of the unfolding test and proves the other component to be pre-tectonic. The Atienza andesites have been collected at six sites from at least four petrographically different units (Van der Voo, 1967). The 33 samples yielded after elimination of the small secondary component very consistent "characteristic" directions. We adopt here the term "characteristic magnetization" for this component, because conclusive proof of the primary character of the residual component revealed by progressive demagnetization is lacking (no unfolding test). When we find the same residual component in several samples from a given site, we speak of the characteristic magnetization of that site. The same concept can be extended to a regional scale (Gregor and Zijderveld, 1964). An Upper Silurian age has been attributed to these Atienza Andesites (Schroeder, 1930). Schroeder described the volcanics as andesitic flows, covering well-dated Upper Silurian shales. The overlying similar shales, however, are of unknown age. This sequence has a moderate dip caused by the Hercynian orogeny (Schroeder, 1930; Van der Voo, 1967). The possibility remains, therefore, that these andesites have an age between Late Silurian and Late Carboniferous.

New results

To check these data, a few samples from two sites of Late Ordovician age from central northern Portugal were tentatively investigated (Table I, no.1).

Lying on the Precambrian basement, the Ordovician and Silurian sequence forms a long narrow synclinal structure of Caledonian or Early Hercynian folding from Luso to Penacova. Both Precambrian and Silurian are nonconformably overlain by Upper Carboniferous-Lower Permian sandstones. A geologic map with sampling localities indicated is given in Fig.2, compiled after the map by Nery Delgado (Carrington da Costa, 1950).

Samples of un-weathered rock could be collected from two sites of basalt flows, intercalated between unmetamorfosed sediments and tuffs. The initial measurements of the N.R.M. made it clear that the samples from site F, however, were too weakly magnetized ($1 \cdot 10^{-7}$ e.m.u./cm^3).

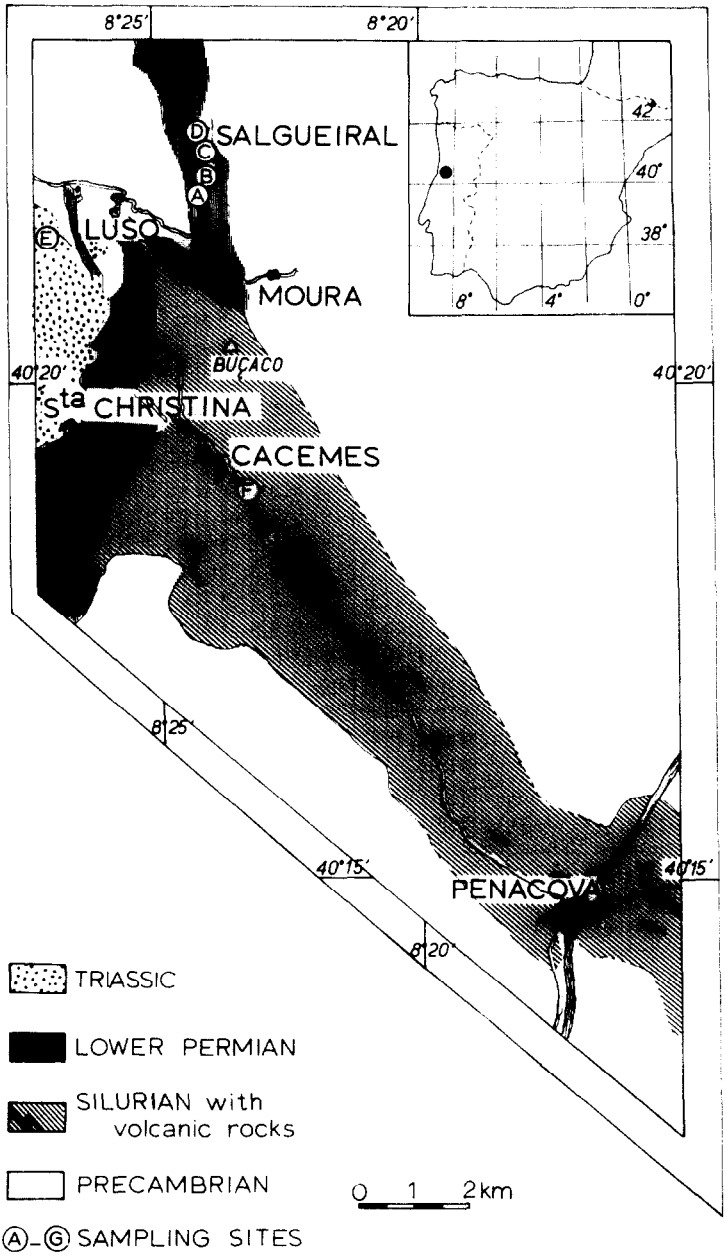


Fig.2. Geologic map of the Buçaco region (Coimbra, central–northern Portugal). The sampling sites are indicated. The map has been compiled after the map by J.F. Nery Delgado, edited and published by Carrington da Costa (1950).

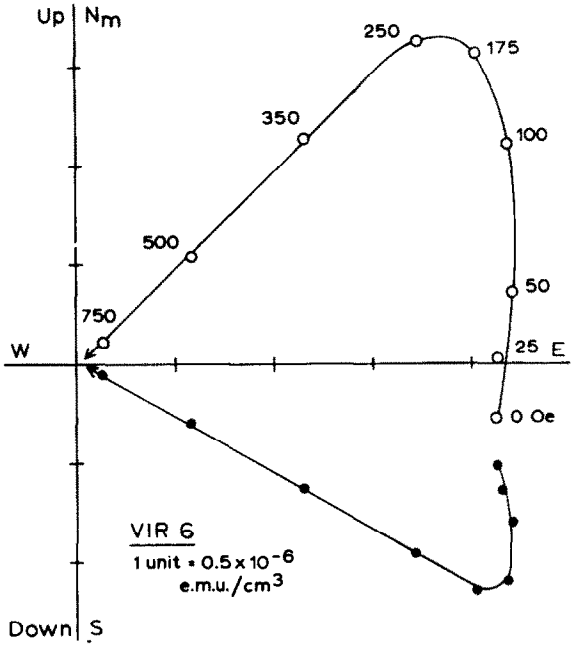


Fig.3. Diagram of a progressive a.c. magnetic field demagnetization of an Ordovician basalt sample from the Buçaco region (site G, Fig.2).

Plotted points represent successive positions - in orthogonal projection - of the end point of the magnetic vector. Full symbols represent projections on the horizontal plane; open symbols represent projections on the east–west vertical plane. Numbers denoted a.c. magnetic field intensities in Oersteds (Oe).

Not corrected for the present-day geomagnetic declination (July 1966: 7.5°W). Nm = magnetic north.

The two samples from site G had intensities of about $1 \cdot 10^{-6}$ e.m.u./cm^3. They have been progressively demagnetized with the aid of a.c. magnetic fields. An example is given by the diagram of Fig.3. Their mean direction of magnetization after tectonic correction is given by $D = 101°$ and $I = +16.5°$ and this does not deviate much from the mean directions ($D = 121°$ and $I = +20°$; $D = 140°$ and $I = +25°$) of the previous results from Almadén.

Devonian and Lower Carboniferous

In southwestern Spain and Portugal two investigations were planned, but no reliable paleomagnetic results could be obtained (Table I, no.4,5). Thirty-eight samples of the following formations were collected.

(*1*) The Devonian volcanic rocks near Almadén (province of Ciudad Real).

(*2*) The Devonian and Carboniferous volcanic rocks near Pueblo de Guzmán (province of Huelva, Spain) and Pomarao (province of Baixo Alentejo, Portugal). The geology of these regions has been described by Doetsch (1953), Almela and Febrel (1960) and Van den Boogaard (1967), respectively. With exception of those from one site (G), all of the samples had but little magnetic remanence, with intensities of about $5 \cdot 10^{-8}$ e.m.u./cm^3. They appeared to have a high susceptibility. The samples, therefore, are not suitable for the necessary demagnetization treatment. Seven samples from site *G*, near Pueblo de Guzmán (province of Huelva, Spain; Fig.7) had higher intensities varying between $10 \cdot 10^{-6}$ and $200 \cdot 10^{-6}$ e.m.u./cm^3, but this material was collected from a dike or sill of which neither the age nor the tectonic position could be reliably determined.

Upper Carboniferous and Lower Permian

It has been pointed out already that the main purpose of the present study is to compare paleomagnetic results from the Iberian Peninsula with those from "stable" Europe; here one has to bear in mind that by "stable" I mean the part of Europe, situated north of the Alpine fold belt, e.g., the Meso-Europe of Stille.

The structural relationships for the Permian, Mesozoic and Tertiary Periods are relatively simple, since they have been influenced by only one (Alpine) orogeny; in the Iberian Peninsula one may distinguish between the stable block of the Iberian Meseta (Fig.1) and the Alpine chains: the Betic Cordillera in the southeast, the Catalanides in the northeast, and the Pyrenees in the north. At the French side of the Pyrenees an important feature is formed by an east–west trending fault zone, the north Pyrenean fault (De Sitter, 1965).

Previous investigations

Recently Van Dongen (1967) has published his results from a paleomagnetic investigation on probable Lower Permian volcanics in the eastern Pyrenees (Table I, nr.9). Van Dongen investigated about 40 samples from ten sites and found a mean direction of magnetization with $D = 169.5°$ and $I = -3°$, after correction for the sometimes considerable dip of the strata.

Introduction to the new results

However, to get a better idea of the relationship between the stable shield of Europe and the shield of the Spanish Meseta, it is of course necessary to obtain information from central Spain and Portugal, since Alpine orogenetic movements have had their influence upon the formations of the Pyrenees.

For this purpose samples were collected from three Upper Carboniferous–Lower Permian rock units:

(1) red sandstones from the Buçaco Formation near Coimbra (Table I, no.6); (2) fine- and coarse-grained red sandstones from the Viar Basin near Sevilla (Table I, no.7); and (3) dikes and sills, which intruded into these sandstones (Table I, no.8).

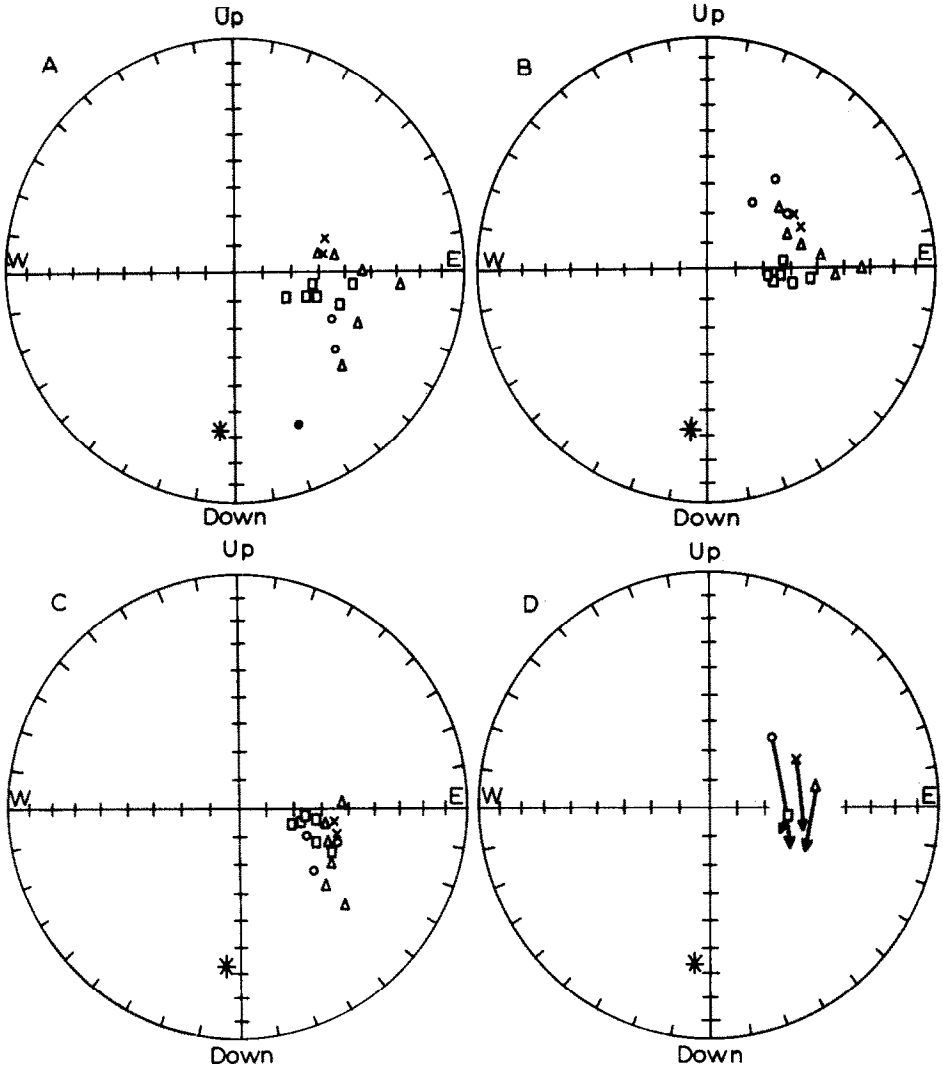


Fig.4. Equal-area projections showing the directions of the magnetic vectors of the Upper Carboniferous–Lower Permian samples from the Buçaco region (Portugal, see Fig.2). The usual (horizontal) equal-area projection, used throughout this paper, would give here an undistinct picture of the measured directions. Therefore all projections are on the east–west vertical plane, full (open) symbols representing north-seeking poles pointing towards the northern (southern) hemisphere. The asterisk denotes the present-day local geomagnetic field direction. The symbols (triangle, square, cross, circle) represent the four different sites.

A. Initially measured total N.R.M., before tectonic correction.

B. The directions of the primary magnetizations, before tectonic correction.

C. The same, after tectonic correction.

D. The mean directions from the four sites, before and after tectonic correction.

The geology of the Coimbra–Luso region (northern central Portugal)

A well-known formation of Late Carboniferous–Early Permian age in northern central Portugal is the Buçaco Formation (Table I, no.6). It has been described a.o. by Texeira (1945) and an excellent geologic map was compiled by Nery Delgado, just before he died in 1908, which was later edited and published (Fig.2) by Carrington da Costa (1950).

Apart from the Ordovician flows (see p.11) and an exposure of Triassic sandstone, four sites of light-purple fine-grained Buçaco Sandstones were visited and seventeen samples were collected. The sandstones non-conformably cover part of the Precambrian and a syncline of Ordovician and Silurian sediments and volcanics. Their age has been determined as Stephano–Autunian[1] on the basis of abundant fossil plants including *Callipteris conferta* Sternb. (Texeira, 1945). The Hercynian orogeny has folded the Buçaco Formation probably in the Lower Permian and after a considerable period of erosion, Upper Triassic (Rhetian) sandstones were non-conformably deposited.

Results from the Coimbra-Luso samples (Portugal)

The measurements initially showed a rather marked grouping of the directions of magnetization (Fig.4A) and revealed intensities between $5 \cdot 10^{-6}$ and $20 \cdot 10^{-6}$ e.m.u./cm^3. The Q-values were about 1.5.

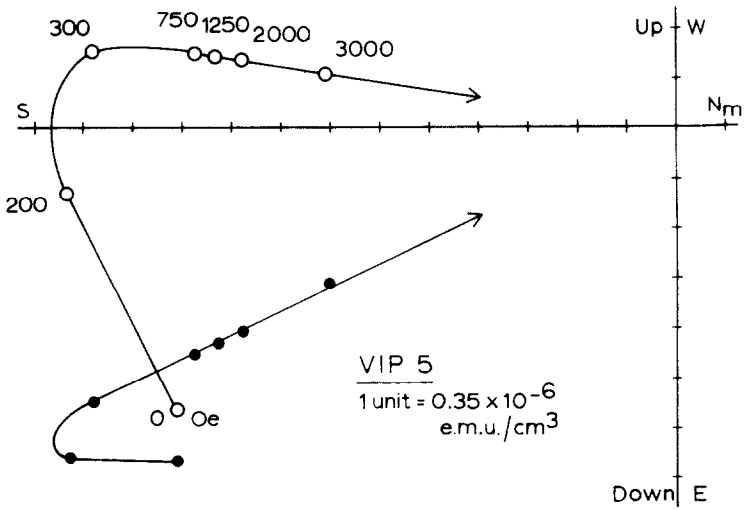


Fig.5. Diagram of a progressive a.c. magnetic field demagnetization of a red sandstone sample from the Buçaco region (Fig.2, site B).

For further explanation see Fig.3. Not corrected for the present-day local geomagnetic declination (August 1967: 9.5°W).

[1]Here I use the name "Stephano–Autunian" for a distinct stage, according to the usage of most modern french stratigraphic geologists. This stage has been defined as the time-span formerly indicated by "Autunien inférieure" and "étage ambigu Permo-Carbonifere" (Grand'Eury, 1890). Doubinger (1956), in her extensive study of the boundary between Permian and Carboniferous, preferred the name "Stephanian D", which is misleading because of the occurrence of *Callipteris* in these formations.

Five samples have been progressively demagnetized in a.c. magnetic fields up to 3,000 Oe (peak value). Two samples have been treated by thermal demagnetization up to 670°C. An example of the a.c. magnetic field demagnetization is given in Fig.5. All samples tested in this way behaved in a similar way: in the a.c. magnetic field trajectory between 0 and 1,000 Oe a secondary component is eliminated, and the higher alternating fields show the decrease of the magnetic vector to pass as a straight line towards the origin.

Consequently, the remainder of the samples has been treated in a few steps in a partial progressive a.c. magnetic field demagnetization. The measurements made in this trajectory (between 1,000 and 3,000 Oe) have been plotted in one graph, after the various corrections for the tectonic dip of the strata were applied (Fig.6). This shows that the directions of these

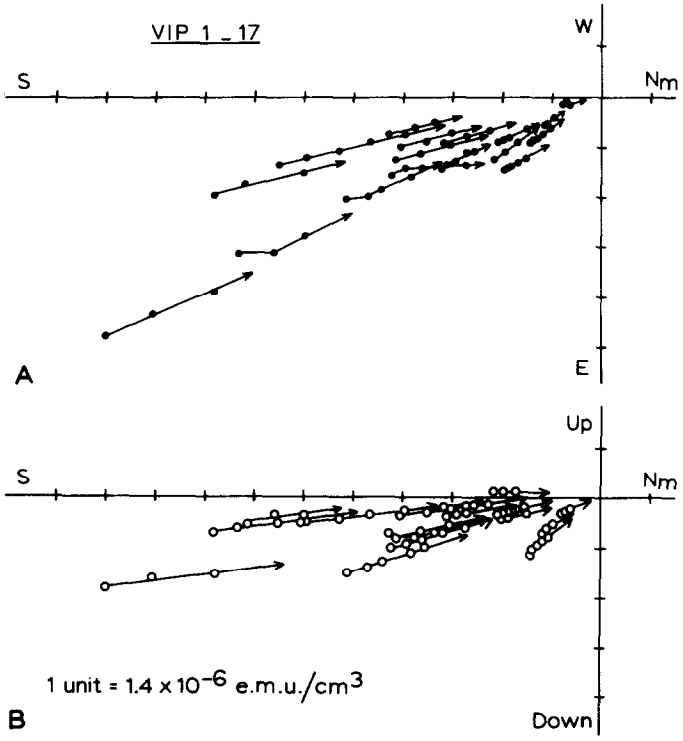


Fig.6. Diagram of the partial progressive demagnetization of all red sandstone samples from the Buçaco region (see Fig.2). A. Horizontal projection. B. Vertical projection.

Plotted points (explanation see Fig.3) for each sample are connected and represent the measurements made in the a.c. magnetic field trajectory where only one component was being eliminated, after correction for the tectonic dip of the strata.

The intensities of the a.c. magnetic fields for the steps of these partial demagnetizations varied between 750 and 3,000 Oe.

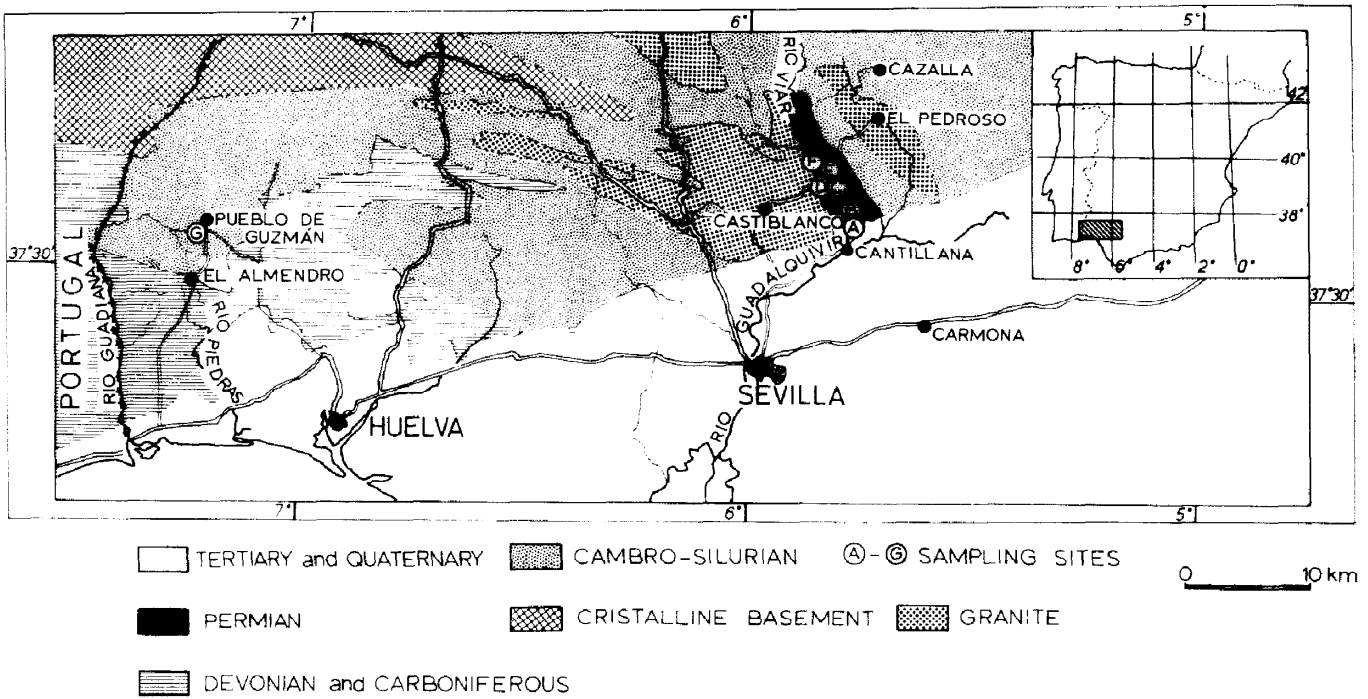


Fig.7. Geologic map of the southern margin of the Spanish Meseta between Sevilla and Huelva. Compiled after the Geologic Map of Spain (1936). The sampling sites are indicated.

hard magnetizations become similar after unfolding. To illustrate this, the directions of the magnetizations and the mean directions for each site have been plotted in Fig.4 before and after unfolding. Applying the corrections for the tectonic dip on the mean directions of the four sites, Fisher's α_{95} decreases from 15° to 7° and the precision parameter k (Fisher, 1953) increases from 40 to 180[1]. Consequently, the magnetization must have been acquired before the folding took place.

To conclude, the age of the redbeds has been determined by plant remains as Stephano–Autunian (viz., 285–270 m.y. according to the time-scale of the Holmes' Symposium, 1964, (quoted by International Union of Geological Sciences, 1967), whereas the unfolding test proves that the acquisition of the remanence of these redbeds took place before the subsequent folding in the upper half of the Autunian. Giving unit weight in the statistical analysis a mean direction of magnetization can be computed with $D = 149°$, $I = +11°$ and $\alpha_{95} = 7°$, while when giving unit weight to samples these values become $D = 149°$, $I = +11°$ and $\alpha_{95} = 5°$.

Geology of the Sevilla region (southern Spain)

The geology of the Viar Basin near Sevilla has been described by Simon (1940). The river Viar, a tributary to the Guadalquivir, here runs from north to south in a topographic as well as a geologic basin of flat-lying red sandstones and conglomerates (Fig.7). These redbeds cover granite and Hercynian-folded Paleozoic sediments with a north-northwest–south-southeast trend. Simon (1940) lithologically divided the redbed sequence into an upper and a lower part and determined the age of the lower part as Late Stephanian or Stephano–Autunian (see footnote on p.16) on the basis of fossil plant remains.

The samples were collected from the upper part of the redbeds, consisting of brick-red fine-grained sandstones and purple coarse-grained sandstones with intercalated conglomerate lenses (Table I, no.7).

In the upper Viar beds a sill and some dikes have been reported by Simon (1940) and García de Figuerola (1959). These exposures have been visited, but only one site appeared sufficiently unweathered (site E, Fig.7). Related volcanism, however, has been very important in the whole area and at two sites of dikes in the granite body more samples have been collected. The Viar redbeds, overlying this granite, have not been folded and show a subhorizontal attitude.

Results from the Sevilla samples

Two types of sediments were collected from the Viar redbeds: brick-red fine-grained sandstone and purple fine-grained conglomerate (sometimes almost coarse-grained sandstone). Both groups had intensities varying between $3 \cdot 10^{-6}$ and $8 \cdot 10^{-6}$ e.m.u./cm^3.

The initial measurements on the total N.R.M. immediately revealed a difference in direction between the coarse- grained sandstones on the one hand and the fine-grained sandstones on the other hand (Fig.8A,B). Their

[1]According to McElhinny (1964), who analyzed the statistics of the unfolding test, the unfolding is significant on the 95%-level, when the k_2/k_1 ratio exceeds 4.28 in the case of four units, where k_2 and k_1 are the precision parameters ($k = (N-1)/(N-R)$), after and before unfolding, respectively.

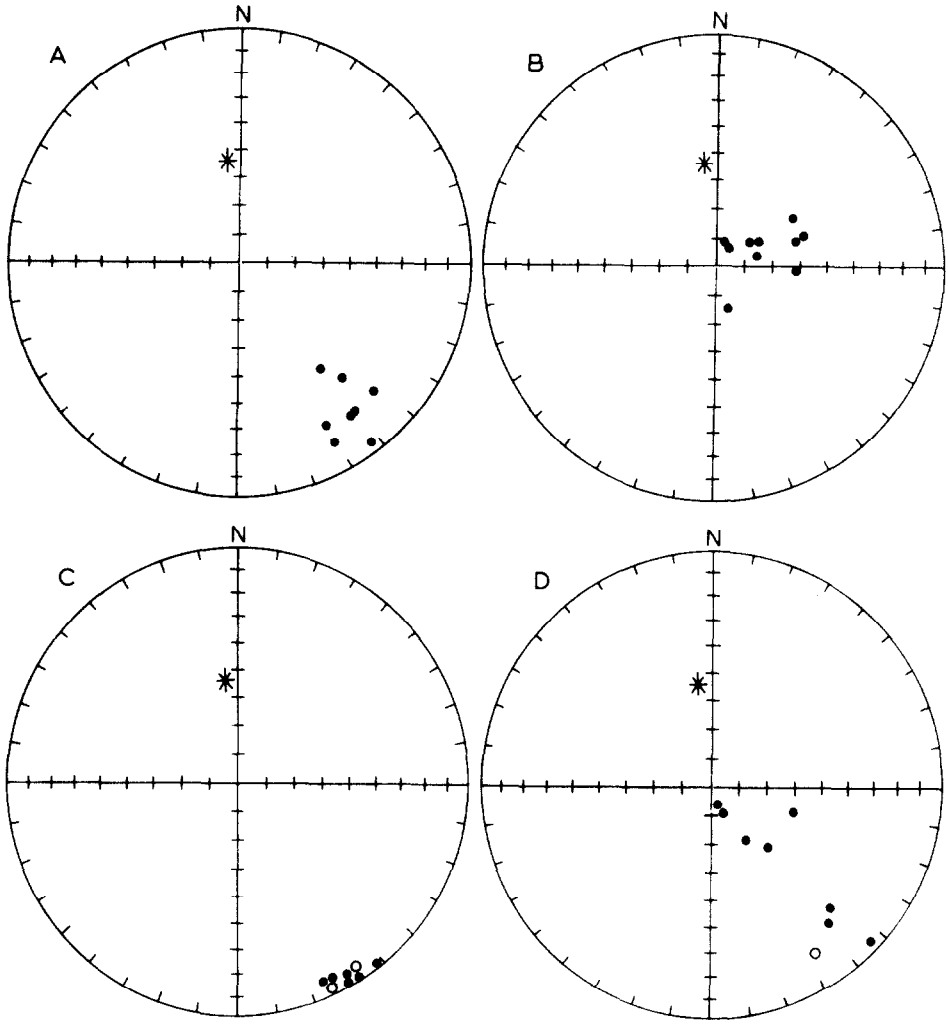


Fig.8. Equal-area projections of the directions of the magnetic vectors in the samples from the Viar redbeds (Fig.7).

A. Initially measured (total) N.R.M. of the coarse-grained sandstones and conglomeratic samples, before tectonic correction.

B. Initially measured (total) N.R.M. of the fine-grained sandstone samples, before tectonic correction.

C. The characteristic components in the coarse-grained sandstones and the conglomeratic samples, after tectonic correction.

D. Directions of the magnetic components after "cleaning" with the aid of thermal demagnetization up to 660°C, obtained from the fine-grained sandstone samples after tectonic correction.

Open (full) symbols represent north-seeking poles pointing upwards (downwards), with a negative (positive) inclination. The asterisk represents the present-day local geomagnetic field direction.

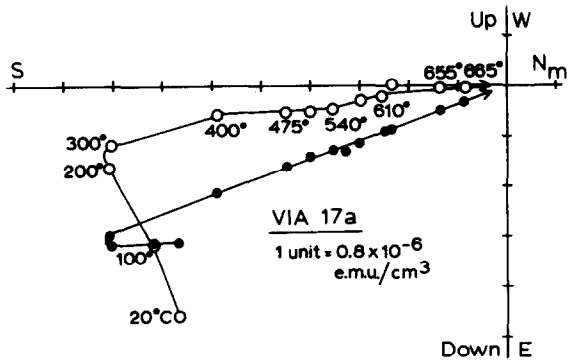


Fig.9. Diagram of a progressive thermal demagnetization of a coarse-grained sandstone sample from the Viar redbeds (Fig.7). Not corrected for the present-day geomagnetic declination (August 1967: 8°W). For further explanation (symbols) see Fig.3.

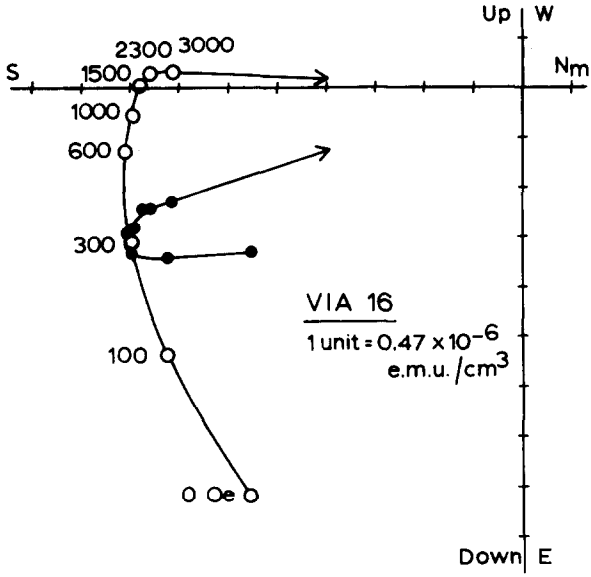


Fig.10. Diagram of a progressive a.c. magnetic field demagnetization of a coarse-grained sandstone sample from the Viar redbeds (Fig.7). Not corrected for the present-day geomagnetic declination (August 1967: 8°W). For further explanation see Fig.3.

total N.R.M.'s have then been analysed both by a.c. magnetic field and thermal demagnetization. In Fig.9 a diagram is presented of a thermal demagnetization of a coarse-grained sandstone sample. In this sample, between room temperature and 300°C, a magnetic component is eliminated which has a direction conformable to the present-day local geomagnetic field and thus is likely to be secondary. At higher temperatures the decrease

of the magnetic vector passes through the origin as a straight line, revealing that now only one component was present in the sample. A.c. magnetic field demagnetization yielded similar information about the other conglomeratic samples (Fig.10). There proved to be no significant difference between the directions of the characteristic magnetizations, as obtained from thermal and a.c. magnetic field demagnetization, respectively. The characteristic directions of magnetization of the conglomerates are plotted in equal area projection in Fig.8C. One can see that the directions of all samples are consistent after demagnetization. A mean direction of magnetization has been computed, giving unit weight to sites in the statistical analysis: $D = 151°$, $I = +2°$ and $\alpha_{95} = 6°$ ($N = 3$). Giving unit weight to samples in the analysis, α_{95} becomes 4.5° ($N = 8$).

The N.R.M. of the samples of the fine-grained sandstone series, however, was far more difficult to interpret. In Fig.11, both thermal and

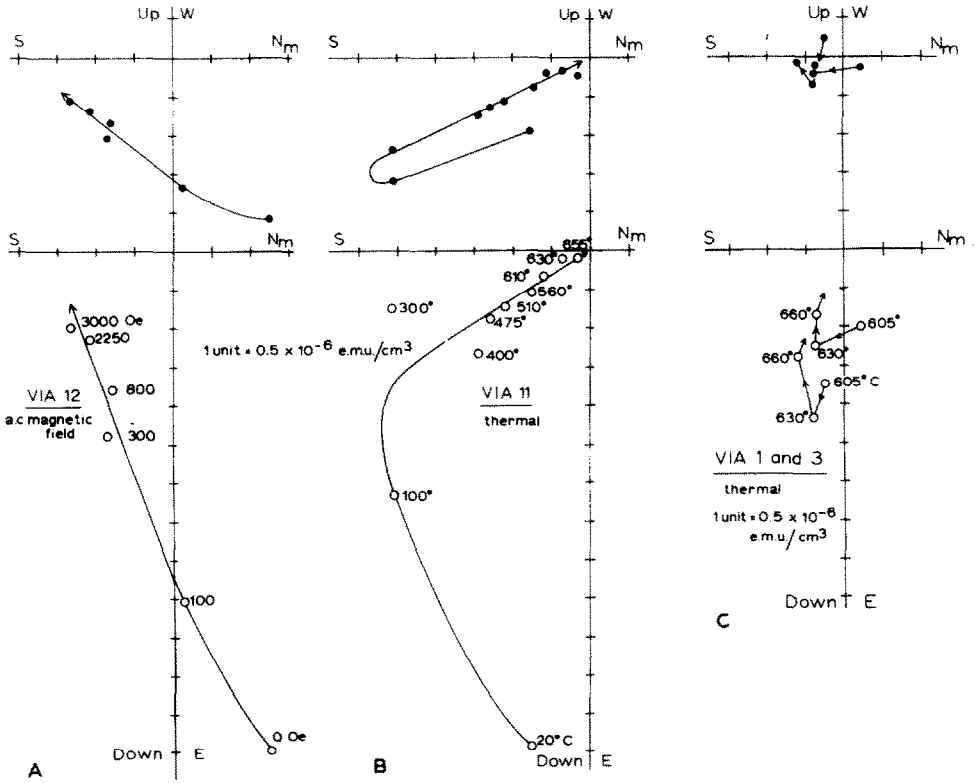


Fig.11. Demagnetization diagrams of some fine-grained sandstone samples from the Viar redbeds (Fig.7).

A. Progressive a.c. magnetic field demagnetization of sample VIA 12 up to 3,000 Oe.

B. Progressive thermal demagnetization of sample VIA 11 up to 655°C.

C. Partial progressive thermal demagnetizations of samples VIA 1 and 3 up to 660°C.

For explanation of the symbols see Fig.3.

a.c. magnetic field demagnetization diagrams are given. They show clearly that the influence of the secondary magnetization was much greater here. A.c. magnetic fields up to 3,000 Oe (peak value) did not succeed in eliminating these secondary components (Fig.11A). In thermal demagnetization, moreover, it appeared difficult to measure the small amount of "harder" magnetization, since this component had intensities as low as $0.5 \cdot 10^{-6}$ e.m.u./cm^3. The large noise/signal ratio here causes a considerable scatter (Fig.11B). All remaining fine-grained sandstone samples have been treated in thermal demagnetization with temperatures of 600°, 630° and 660°C (see Fig.11C). In Fig.11C the demagnetization diagrams not only display the just mentioned scatter, they also reveal that the secondary and characteristic magnetizations both are being eliminated at the same time: the decrease of the magnetic vector does not take place in a direction towards the origin.

When one compares the directions, computed after 660°C (plotted in Fig.8D) with those of the conglomerates and coarse-grained sandstones (Fig.8C), it is once more obvious that in the fine-grained sandstone samples the secondary magnetizations have not been eliminated completely as there remains a definite "streaking" towards the present-day local geomagnetic field direction.

The magnetic behaviour of these fine-grained sandstones thus proves to be different from that of the coarse-grained sandstones and the conglomerates. It followed from the demagnetizations that for the friable fine-grained sandstones both the secondary and the characteristic magnetizations were eliminated in the same trajectory and that, unfortunately, the ratio of these magnetizations was rather high. This might perhaps be due to the permeability, which permitted groundwater to circulate and to cause a chemical remagnetization. The harder conglomerates and coarse-grained sandstones, on the other hand, are much more cemented with a dense purple matrix. Ore microscopy revealed abundant fine-grained hematite to be present in the matrix and a few larger grains of hematite as well, with a diameter up to 0.4 mm. The small fragments of quartz and crystalline rock (diameter up to 6 mm) contained no hematite.

From three sites of dikes and sills, intruding the granite and the Viar series, eight samples were collected for comparison (Table I, no.8). A.c. magnetic field demagnetization (Fig.12C) revealed an extremely "hard" magnetization to be present in the samples, which is probably due to hematite found present by ore microscopy.

To be sure that the secondary magnetizations were fully eliminated, thermal demagnetization was required for all samples and consequently 14 cores (specimen), drilled from the hand-samples embedded in paraffin, were heated. An example of a progressive thermal demagnetization is given in Fig.12A. The other specimen were heated to 600°, 630° and 660°C, respectively (Fig.12B).

The directions became well grouped after demagnetization, as is shown in Fig.13. From these directions a mean direction of magnetization has been computed. Giving unit weight to specimen these values become: $D = 155.5°$, $I = +13°$ and $\alpha_{95} = 4°$ ($N = 14$), giving unit weight to sites: $D = 155°$, $I = +10.5°$ and $\alpha_{95} = 13°$ ($N = 3$). Since the neighbouring Viar sediments have not been folded, no tectonic correction is needed.

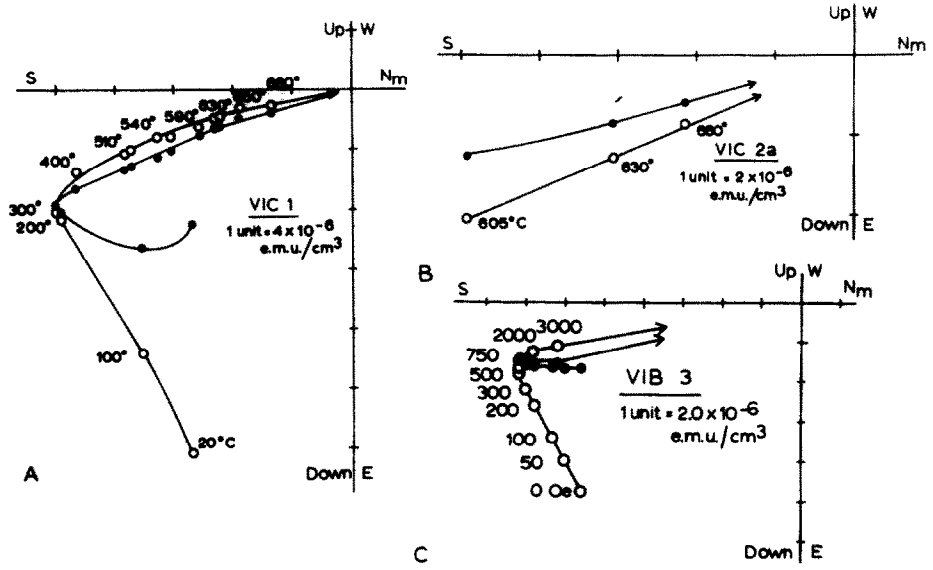


Fig.12. Diagrams of progressive a.c. magnetic field and thermal demagnetizations of three samples from the dikes and sill in the Viar region (sites D and E, Fig.7). For explanation of the symbols see Fig.3.

A. Thermal progressive demagnetization of sample VIC 1 up to 660°C.

B. Partial progressive thermal demagnetization of specimen VIC 2a.

C. Progressive a.c. magnetic field demagnetization of sample VIB 3 up to 3,000 Oe.

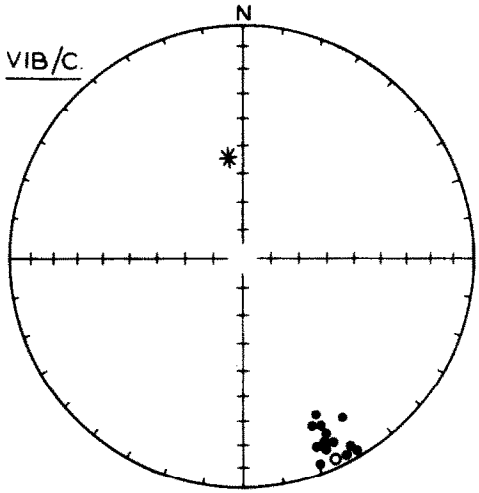


Fig.13. Equal-area projection of the directions of the characteristic components, as revealed by the demagnetization techniques. Samples from the dikes and sill from the Viar region (Fig.7). For further explanation (symbols) see Fig.8.

Permo-Triassic and Triassic

Previous investigations

Several paleomagnetic investigations on Triassic rocks were made in Spain previously, especially in the Spanish Pyrenees. Eight years ago, Van der Lingen (1960) was the first to find stable remanence in Permo-Triassic rocks from the central Pyrenees and a few years later Schwarz (1962) published his results of andesites and redbeds from the adjoining region (Table I, no.11–13). Both found pole positions diverging from those found for stable Europe. Van Dongen (1967) tentatively reported similar results from the Triassic redbeds in the eastern Pyrenees.

In northwest and central Spain (near Vilaviciosa and Alcolea, respectively, Fig.14; Table I, no.18), Triassic redbeds were first sampled by Clegg et al. (1957) who found only present-day geomagnetic field directions to be present in their samples. It is now recognized that these magnetizations were secondary.

The present author published two studies on Triassic redbeds from the Spanish Meseta (Van der Voo, 1967, 1968b; Table I, no.20,17). The first analysed redbed samples from Alcázar de San Juan on the eastern margin of the Meseta. The characteristic magnetizations yielded a mean direction with D = 359.5° and I = + 23°. The second study described the paleomagnetism



Fig.14. Location map of the Triassic exposures on the Iberian Meseta. Sampling localities are indicated by a circle. (Scale 1:2,500.000)

of redbed samples from Atienza, on the northeastern margin of the Meseta. The magnetic directions obtained were in agreement with those found by Clegg and his co-workers (Clegg et al., 1957). Only secondary magnetizations were revealed in the analysis of the N.R.M.

The localities and results from the red sandstones of the margin of the Spanish Meseta

Though previous studies so far were rather disappointing, various other groups of the same continental brick-red sandstones and siltstones were collected in southern Spain and Portugal (Table I, no.14–16). Together with the investigations mentioned above, the positions of all sampling localities on the Meseta, are shown in Fig.14, and one sees that most of the Triassic redbeds occurring on the Meseta have been visited (about 200 samples).

Few words need to be wasted on the results. Though most samples had intensities of up to about $10 \cdot 10^{-6}$ e.m.u./cm^3, all magnetizations appeared to be secondary. As an example the directions measured on the samples of Algarve (southern Portugal) are given in equal area projection in Fig.15.

Geologic situation of the Garralda anticline (western Pyrenees)

A large group of redbed samples has been collected by Mannot (1965) in the southern zone of the western Pyrenees near Garralda (Table I, no.19). The Triassic sequence here consists of an alternation of conglomerates, red arkoses and more massive dark red siltstone layers from which most of the samples were taken. Near Garralda, it non-conformably overlies the Paleozoic, consisting mainly of Devonian limestones and shales. Elsewhere

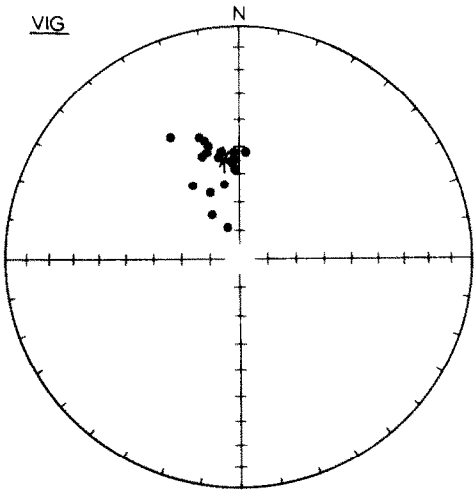


Fig.15. Equal-area projection of the directions of magnetization in the samples from Algarve (Fig.14, southern Portugal): initial total N.R.M. For explanation of the symbols see Fig.8.

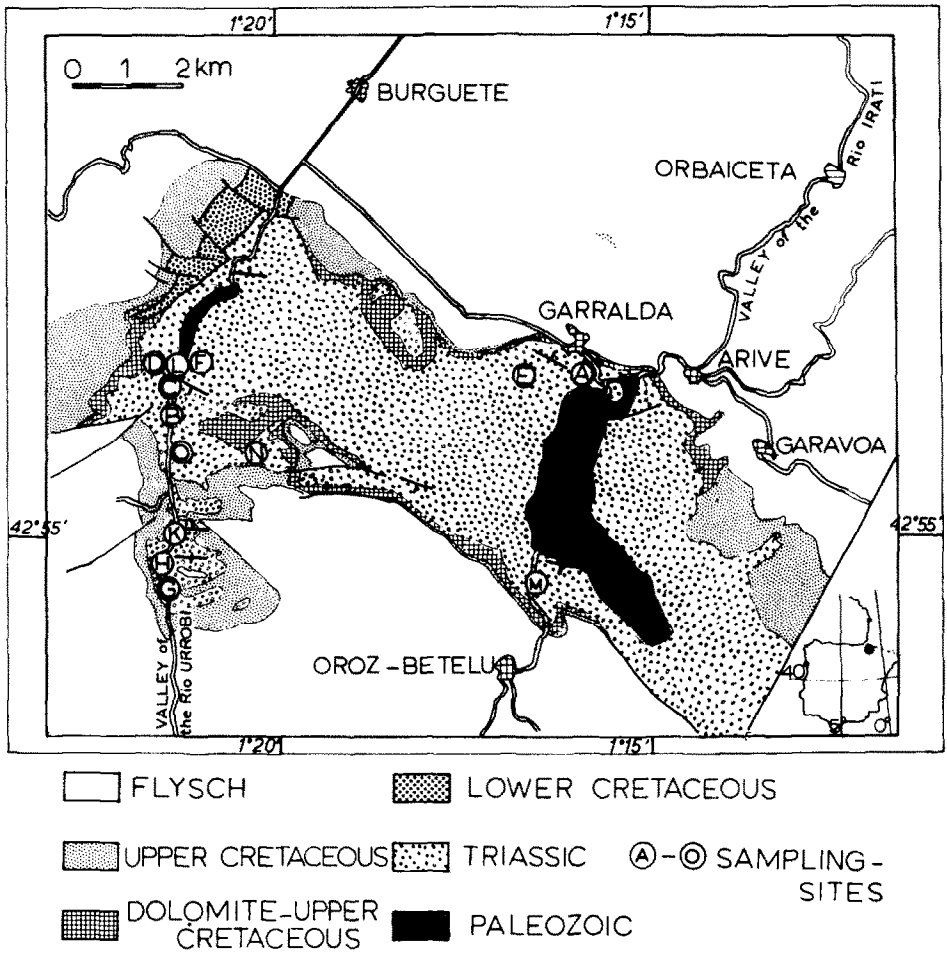


Fig.16. Geologic map of the Garralda region (western Pyrenees). The sampling sites are indicated.

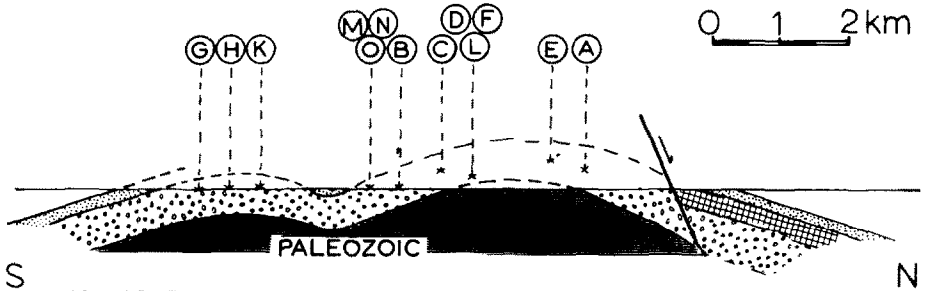


Fig.17. Schematic cross-section through the anticlinal structure of the Garralda region (Fig.16). Projected sampling sites are indicated. Legend see Fig.16.

a homologous unconformity can be found between similar Triassic redbeds and Permian conglomerates. The Triassic section is overlain by Cretaceous and Tertiary sediments. The exact age of the series has not yet been determined, due to the lack of fossils (Ciry et al., 1963).

A detailed geologic map of the Garralda region has not been published. The geologic map of Fig.16 with the sampling sites indicated, has been compiled after detailed mapping by Mannot (1965). The map of Fig.16 shows a large part of the exposures of the Triassic redbeds to form an anticlinal structure with an east–west trend. Two rivers, the Urrobi and the Irati, expose in the nucleus of the anticline the underlying Paleozoic. The sediments surrounding the Triassic are all younger in age, and consist of Cretaceous limestones and dolomites and Lower-Tertiary sediments in flysch facies (Ciry et al., 1963). Due to the occurrence of normal faults (Fig.16) and the considerable overgrowth, it proved to be impossible to correlate the sampling sites stratigraphically. A schematic cross-section based on the observations of Mannot (1965) is presented in Fig.17.

Results of the Garralda samples

About 160 samples were collected; of these a case with 65 samples was lost before arriving in Utrecht. The initial measurements on the total N.R.M. of the remaining 95 samples revealed that the red siltstones were rather weakly magnetized. The intensities varied between $0.5 \cdot 10^{-6}$ and $5 \cdot 10^{-6}$ e.m.u./cm^3, the Q-values generally varied between 0.5 and 3. The initially-measured directions, moreover, showed a considerable scatter with a concentration of directions conforming to the present-day local geomagnetic field (Fig.18), and "streaking" away in two opposite directions, towards the north-northwest and south.

As it was hoped that elimination of secondary magnetizations would

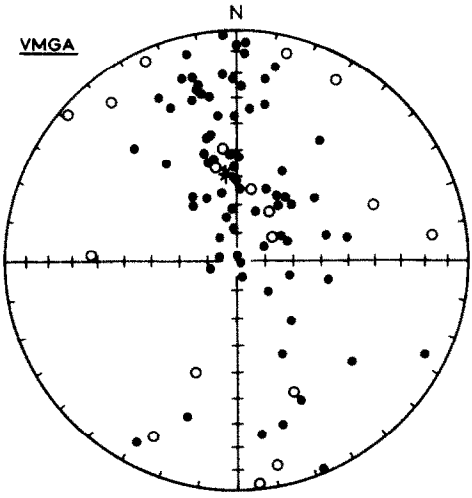


Fig.18. Equal-area projection of the directions of the total N.R.M.'s from the Garralda samples (Fig.16, western Pyrenees). For explanation of the symbols see Fig.8.

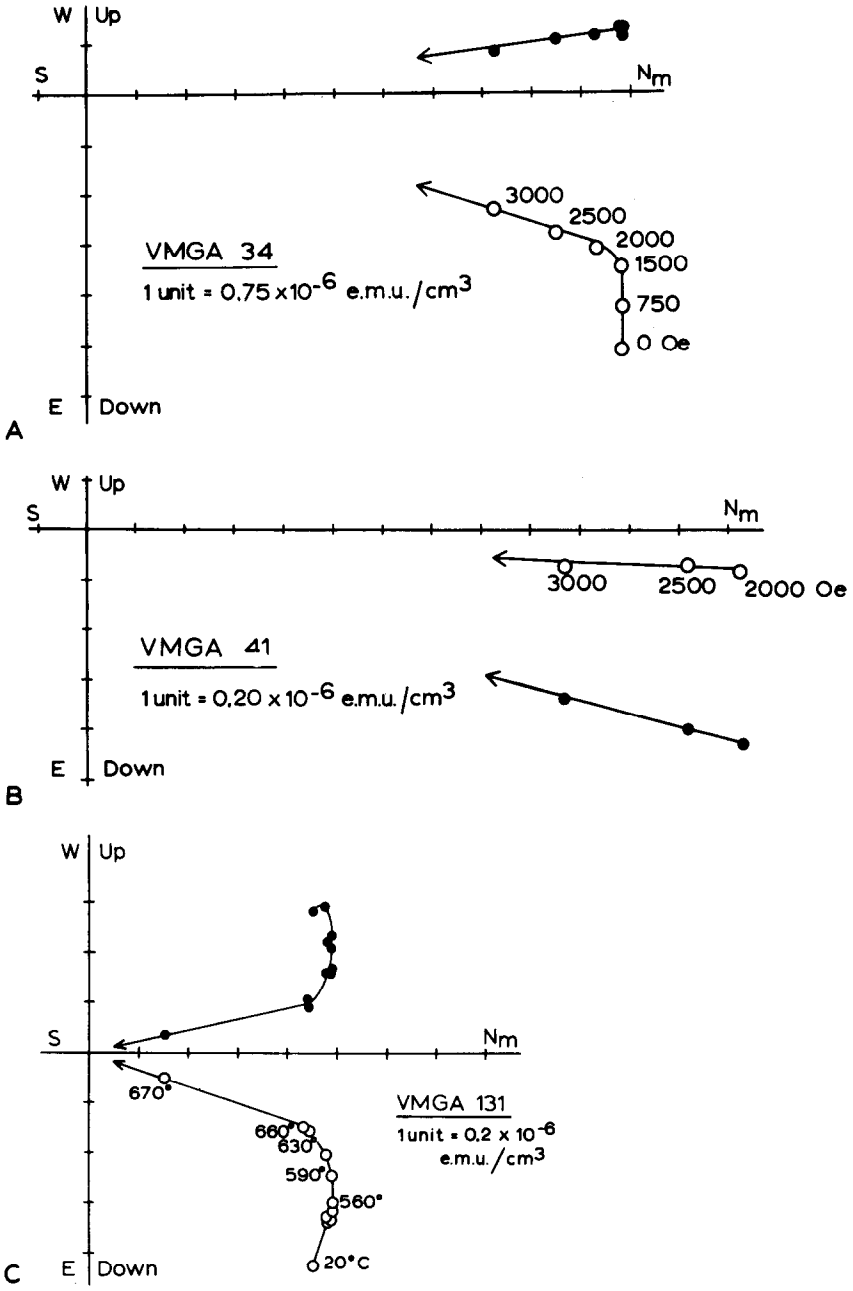


Fig.19. Diagrams of two a.c. magnetic field (A,B) and a thermal (C) demagnetization of Triassic Garralda samples (Fig.16, western Pyrenees).

For explanation of the plotted symbols see Fig.3.

Not corrected for the present-day geomagnetic declination (July 1965: 6.5°W).

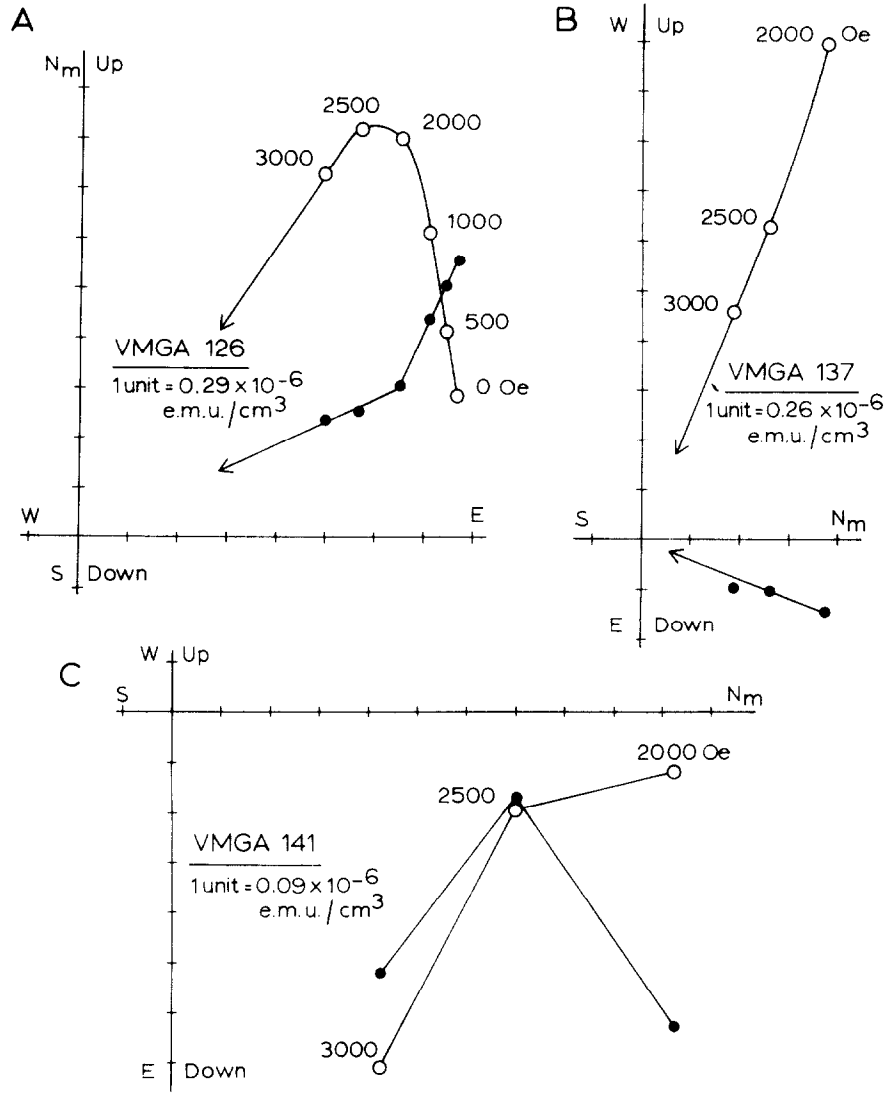


Fig.20. Diagrams of three a.c. magnetic field demagnetizations on Triassic Garralda samples (Fig.16, western Pyrenees). For explanation of the plotted symbols see Fig.3.

A. Progressive demagnetization on sample VMGA 126.

B. Partial progressive demagnetization on sample VMGA 137.

C. Partial progressive demagnetization on an unstable sample (VMGA 141).

Not corrected for the present-day geomagnetic declination (July 1965: 6.5°W).

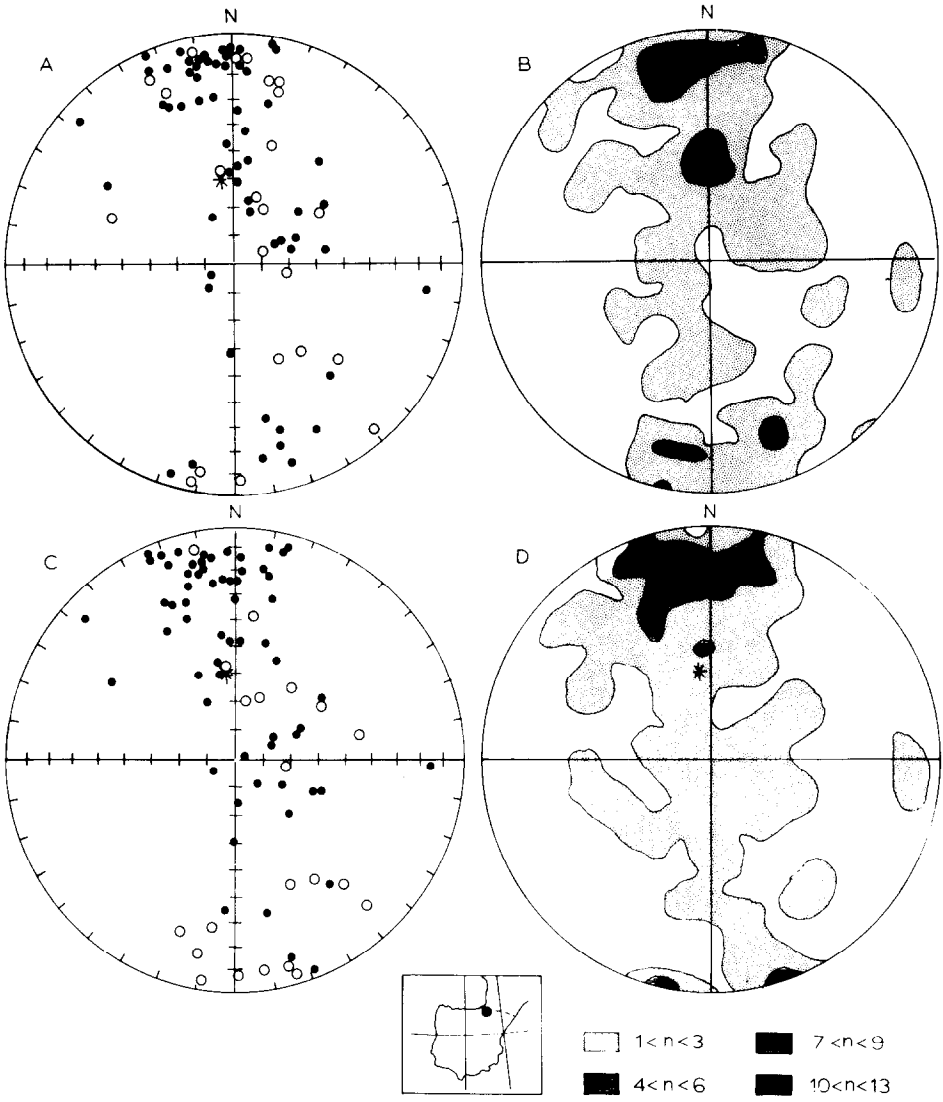


Fig.21. The Triassic Garralda samples (Fig.16, western Pyrenees).

A. Equal-area projection showing the directions of the magnetic vectors revealed by demagnetization techniques, before tectonic correction, and after elimination of the "softer" magnetic components.

B. Density distribution of Fig.21A. This has been realised by plotting all directions of Fig.21A in the lower hemisphere (regardless of polarity) and counting the number of directions in each equal area all over the projection.

C. Equal-area projection showing the directions of Fig.21A after tectonic correction.

D. Density distribution of Fig.21C. For explanation of the procedure see Fig.21B.

improve the concentration of the directions, the N.R.M. of all samples was analyzed with the aid of a.c. magnetic demagnetization. Some examples of progressive and partial progressive demagnetizations are presented in Fig.19 and Fig.20. In Fig.19C, a thermal demagnetization similar in character has been added. It can be observed in these diagrams that two components are present in the samples. The softer component, directed approximately along the recent geomagnetic field in Spain, is assumed to be secondary. These components are eliminated in a.c. magnetic fields up to 2,500 Oe and in temperatures of about 630°C.

The "harder" component generally forms more than 50% of the total N.R.M. The directions of these harder components have been plotted in equal area projection in Fig.21A,C before and after applying the correction for the tectonic dip of the strata.

Some exceptions were made in the case of unstable behaviour, due to viscous magnetizations (see, for example Fig.20C). As it was not possible to determine a reliable direction from these samples, they have been excluded from further analysis (eight samples).

Examination of Fig.21A, showing the projection of the "cleaned" directions, reveals that part of the directions remain concentrated around the present-day geomagnetic field. The directions of the other samples are grouped around a mean direction with an inclination considerably less than the local present-day geomagnetic field and with a north–northwest declination (regardless of polarity).

I have tried to find a representative method to depict the significance of these directions. Therefore, in Fig.21B,D a density distribution has been depicted. This has been realized by plotting the directions regardless of polarity in the lower hemisphere of an equal area projection, and counting

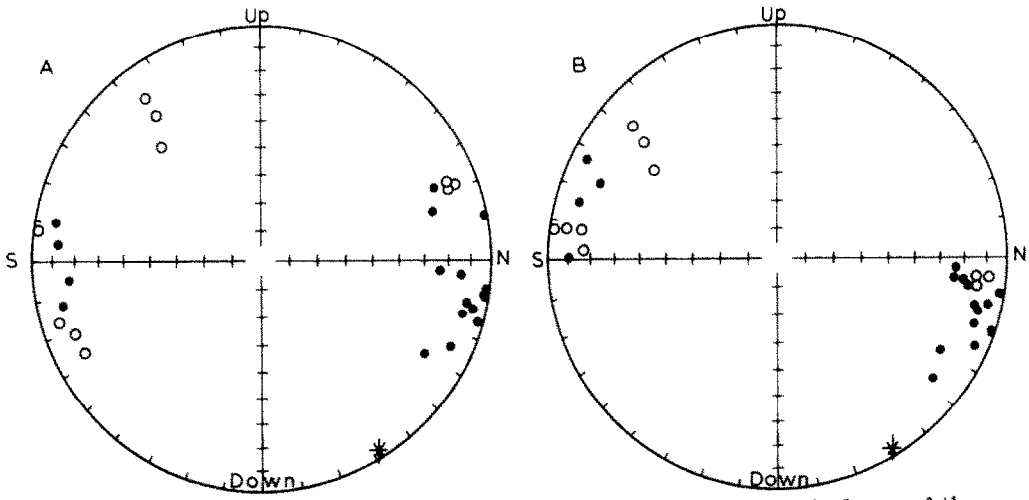


Fig.22. Equal-area projections on the north–south vertical plane, of the characteristic directions of magnetization. Samples from five sites of the Triassic Garralda redbeds. A. before tectonic correction; B. after tectonic correction.

For explanation of the symbols see Fig.4.

the number of directions present in each "equal" area all over the projection. The areas of equal density thus obtained, clearly show the different concentrations.

One difficulty arises: the only way to give the reader full information, for instance on the change in direction owing to the unfolding, is to plot the sample numbers in Fig.21 as well. These numbers, however, would tend to obscure the projections completely. Hence a vertical north–south equal area projection has been given with the directions of five sites only (Fig.22). As can be seen in Fig.17 these sites have various dips (sites M, N and O to the south; site K to the north; site L subhorizontal). After unfolding the directions become well grouped (Fig.22B).

One now may summarize the measurements as follows:

(1) after elimination of the secondary magnetizations 60% of the samples have characteristic directions with normal and reversed stable magnetizations, which become better grouped after unfolding (Fig.22B);

(2) 25% of the samples contained only present-day local geomagnetic field directions. These magnetizations are likely to be secondary;

(3) 15% of the samples contained stable remanent magnetizations with scattered directions, all deviating from the directions mentioned above. This group is responsible for the large area with low density in Fig.21B,D. The demagnetization diagrams of Fig.20A,B reveal that these magnetizations were stable. The decrease of the magnetic vector passes through the origin as a straight line after elimination of the softer secondary magnetization, showing that no other component is present.

The first group most probably represents the Triassic field direction. The scattered directions of the third (small) group remain unexplainable (lightning or chemical remagnetization?).

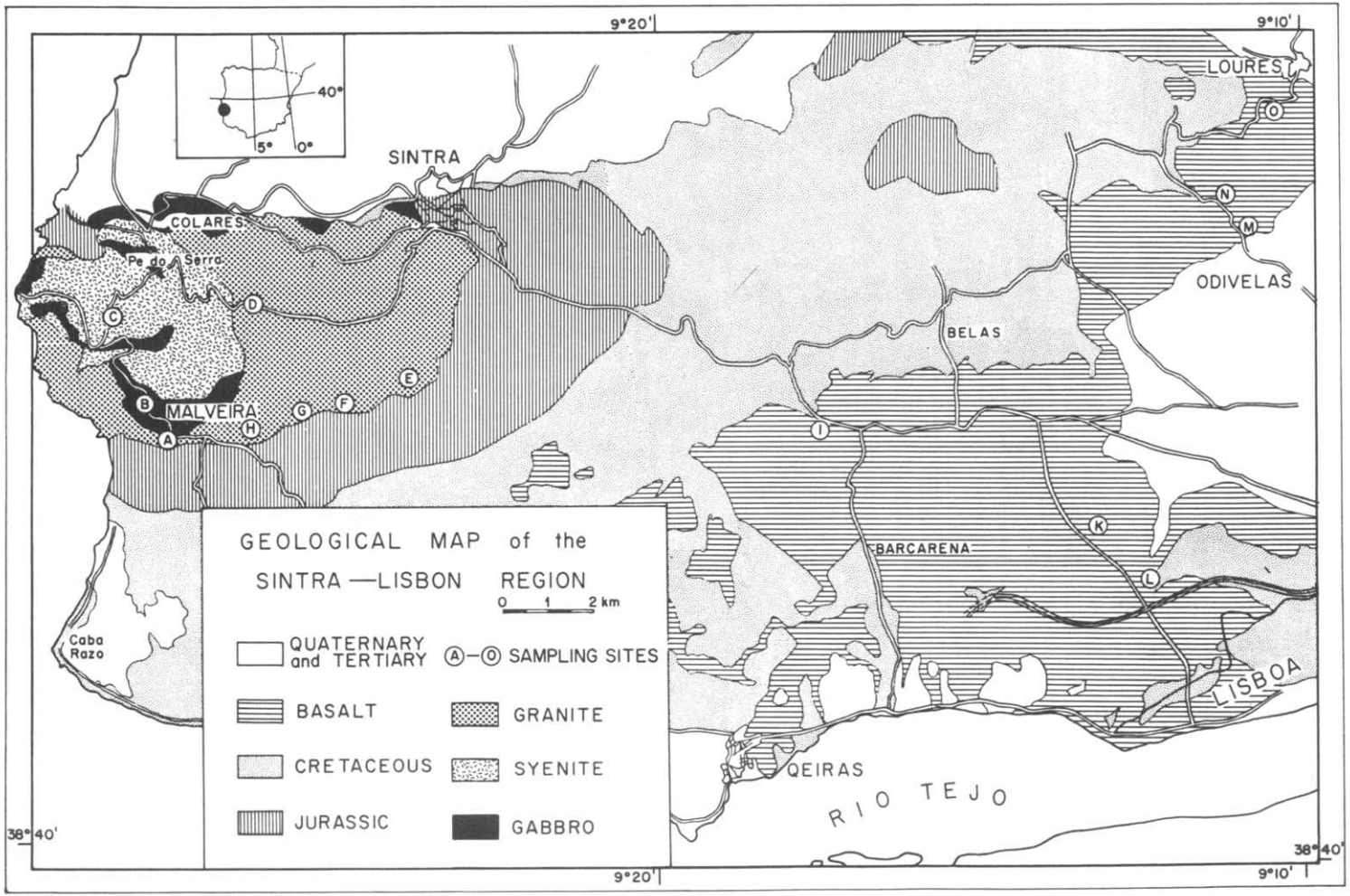
Jurassic

Only a few Jurassic volcanics and sedimentary rocks have been reported from the Spanish Meseta. These rocks seem not suitable for paleomagnetic research, so the only further information on the paleomagnetic history of the Meseta was to come from Cretaceous and Tertiary rocks.

Upper Cretaceous and Eocene

One of the most outstanding regions for collecting volcanic samples is the neighbourhood of Lisbon, where the large intrusive Sintra complex (Upper Cretaceous, Table I, nr.21) is exposed and where basalt flows are intercalated between Upper Cretaceous and Oligocene sediments (Table I, no.22,23). A map of this area with the sampling localities indicated is given in Fig.23.

While this study was in preparation a paper was published by Watkins and Richardson (1968) on the paleomagnetism of the Eocene Lisbon Basalt flows. They reported paleomagnetic properties similar to those that will be described in the following pages, but as Van der Voo (1968c) pointed out in a comment on their paper, they failed to apply the correction for the tectonic dip of the flows.


9°20'
9°10'
40°
5°
0°
SINTRA
LOURES
COLARES
Pe do Serro
ODIVELAS
BELAS
MALVEIRA
BARCARENA
Caba Razo
LISBOA
QEIRAS
RIO TEJO
38°40'
GEOLOGICAL MAP of the SINTRA —LISBON REGION
0 1 2 km
QUATERNARY and TERTIARY
A—O SAMPLING SITES
BASALT
GRANITE
CRETACEOUS
SYENITE
JURASSIC
GABBRO

Geology of the Lisbon region

Radiometric age determinations yielded an age of 80 m.y. for the Sintra Granite (Bonhomme et al., 1961). It has intruded into Jurassic and Cretaceous sediments, which have been distorted and now form a dome-structure around the intrusive body. Twenty-five samples were taken from six sites of granitic rock and two sites of gabbro. It is supposed that no post-intrusive tilting of parts of the granitic complex did occur.

The basalt flows have a moderate tectonic dip towards the east or southeast. Twenty-two samples have been collected in six sites with a different dip each.. For more complete information on the geology, petrography and chemistry of these volcanics the reader is referred to the geologic map of the Lisbon region (1935–1960) and the publications by Zbyszewski (1963,1964) and Watkins and Haggerty (1967).

Analysis of the N.R.M. of the Upper Cretaceous granite

The granitic samples had intensities of total N.R.M. varying between $30 \cdot 10^{-6}$ and $140 \cdot 10^{-6}$ e.m.u./cm^3 and Q-values of about 1. The gabbroid samples had higher intensities (about $2{,}000 \cdot 10^{-6}$ e.m.u./cm^3).

A.c. magnetic field and thermal demagnetization diagrams are given in Fig.24. In fields of 300 Oe or at about 400°C the secondary components are eliminated. Between 300 and 3,000 Oe or at temperatures above 400°C, the characteristic component is found as shown by the diagrams. In most samples at 3,000 Oe the magnetization has been completely eliminated. The directions of these " harder" components have been plotted in Fig.25A. All magnetizations have more or less the same direction, thus being characteristic for the Sintra complex. Giving unit weight to sites a mean direction has been computed with $D = 359°$, $I = +43.5°$ and $\alpha_{95} = 8°$. Giving unit weight to samples the mean direction becomes $D = 359°$, $I = +42.5°$ and $\alpha_{95} = 5°$.

Analysis of the N.R.M. of the Eocene basalts of the Lisbon region

The basalt samples had intensities varying between $2 \cdot 10^{-3}$ and $11 \cdot 10^{-3}$ e.m.u./cm^3 and Q-values between 2 and 10.

As can be seen in the demagnetization diagrams (Fig.26) most samples contained practically one component of magnetization. When a soft secondary magnetization was present it could easily be eliminated in a.c. magnetic fields of 50 Oe. The samples from site N (Odivelas), however, and two samples of site O (Loures) lost 95% of their remanence in a.c. magnetic below 200 Oe (Fig.27). Although the directions of the remaining magnetizations were not very different from the characteristic directions obtained from the other samples, I have discarded these samples from further analysis since these very soft and sometimes viscous magnetizations are always considered as unreliable (Zijderveld, 1967a).

Applying the corrections for the tectonic dip of the strata the directions become well grouped (Fig.25C) around a mean direction $D = 351.5°$,

---

Fig.23. Geologic map of the Lisbon region (Portugal). The sampling sites are indicated, both for the intrusive Sintra complex and for the Eocene basalts.

The map has been compiled after the Geologic Map of Portugal, sheets Lisbon (1950), Loures (1944), Sintra (1960) and Cascais (1935).

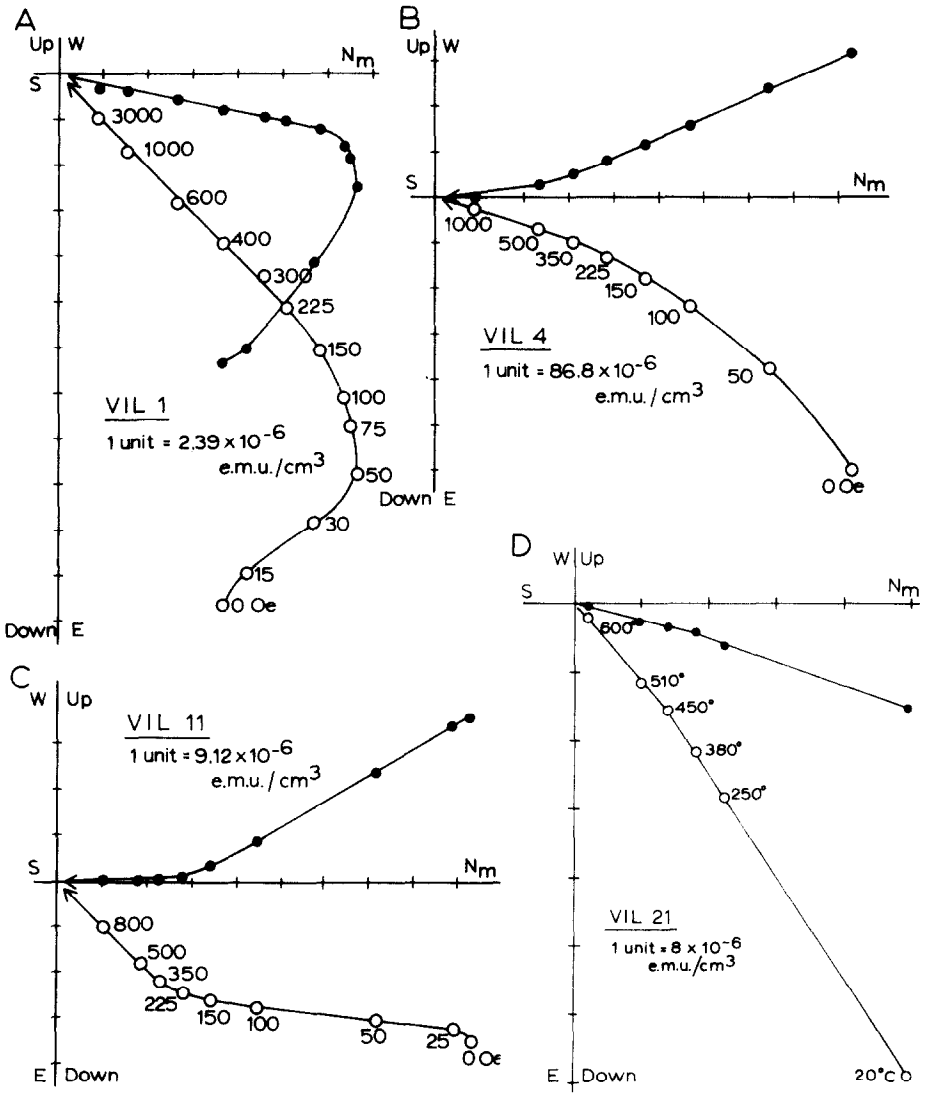


Fig.24. Diagrams of progressive a.c. magnetic field and thermal demagnetizations on the samples from the Sintra complex (Fig.23). For explanation of the symbols see Fig.3.

A. Progressive a.c. magnetic field demagnetization of a granitic sample from site A (VIL 1).

B. Progressive a.c. magnetic field demagnetization of sample VIL 4 from site B (gabbro).

C. Progressive a.c. magnetic field demagnetization of sample VIL 11 from site D (granite).

D. Progressive thermal demagnetization of sample VIL 21 from site G (granite).

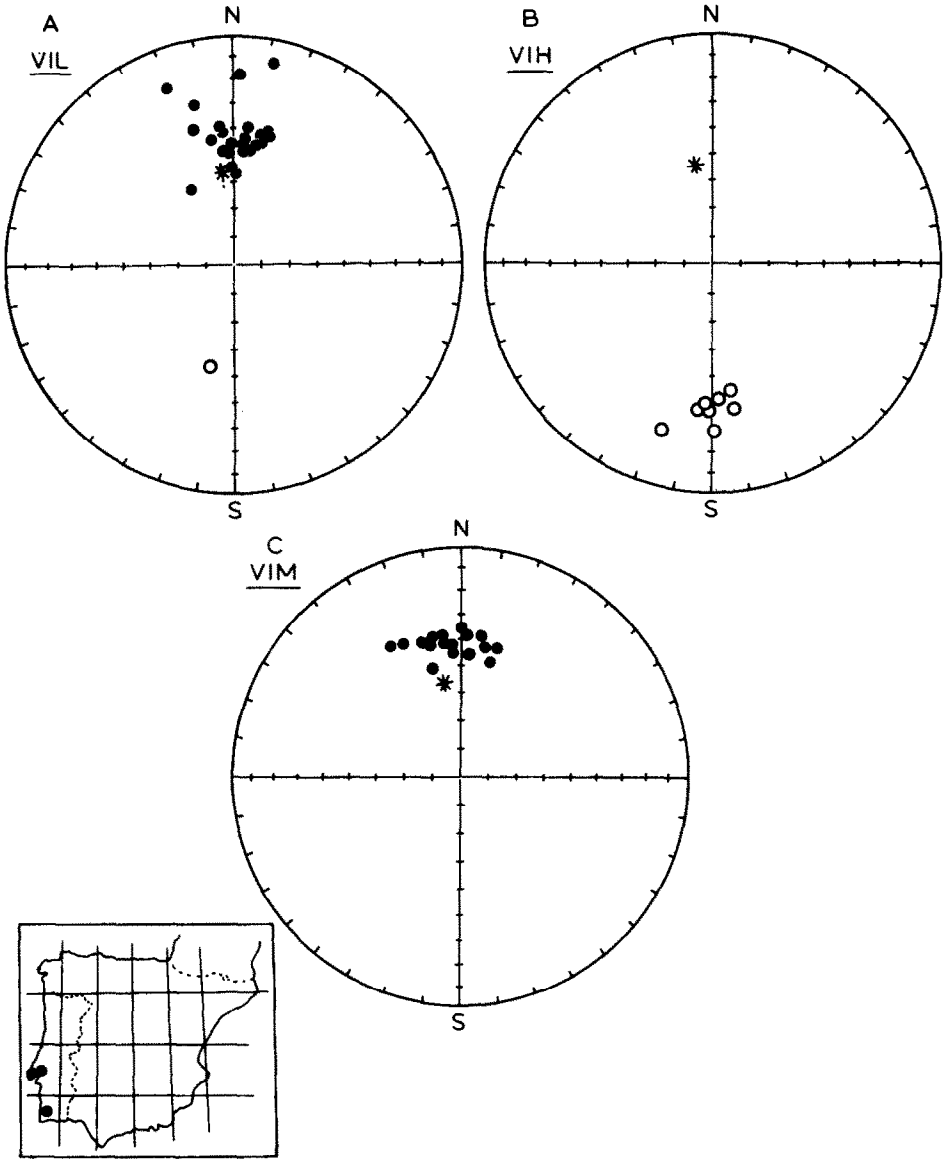


Fig.25. Equal-area projections of the directions of magnetization, after elimination of secondary components.

A. Upper Cretaceous granite. Samples from the intrusive Sintra complex (Fig.23, Portugal).

B. Eocene syenite. Samples from the intrusive syenite complex near Monchique (southern Portugal).

C. Eocene basalts. Samples from the Eocene basalts near Lisbon, after tectonic correction (Fig.23, Portugal).

For explanation of the symbols see Fig.8.

$I = +42°$ and $\alpha_{95} = 10°$, giving unit weight to site means ($N = 5$). The mean direction giving unit weight to samples in the analysis is $D = 354°$, $I = +40.5°$ and $\alpha_{95} = 4.5°$ ($N = 17$). It can be shown for these Eocene basalts that, as in the case of the Lower Permian Buçaco Formation, (p.19), the unfolding test is significant. The k_2/k ratio (3.8) is greater than the confidence limit for $N = 5$ (3.44 according to McElhinny, 1964), where N is the number of units used in the statistical analysis. The correction for the dip of the

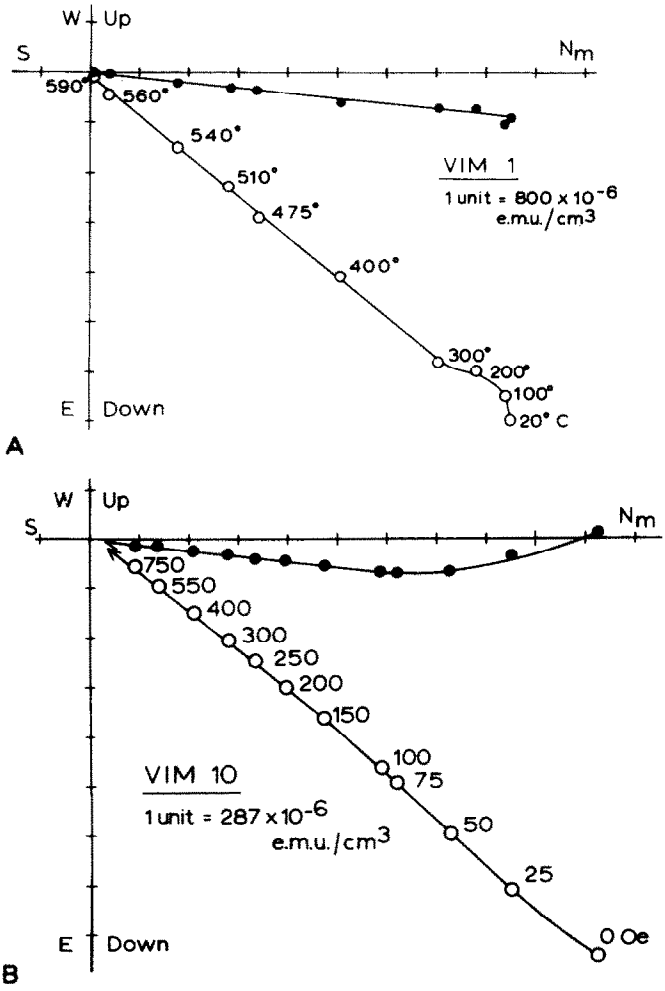


Fig.26. Diagrams of a thermal (A) and a progressive a.c. (B) magnetic field demagnetization of basalt samples from the neighbourhood of Lisbon (Fig.23, Portugal). For explanation of the plotted symbols see Fig.3.

Not corrected for the present-day geomagnetic declination (August 1967: 9.5°W).

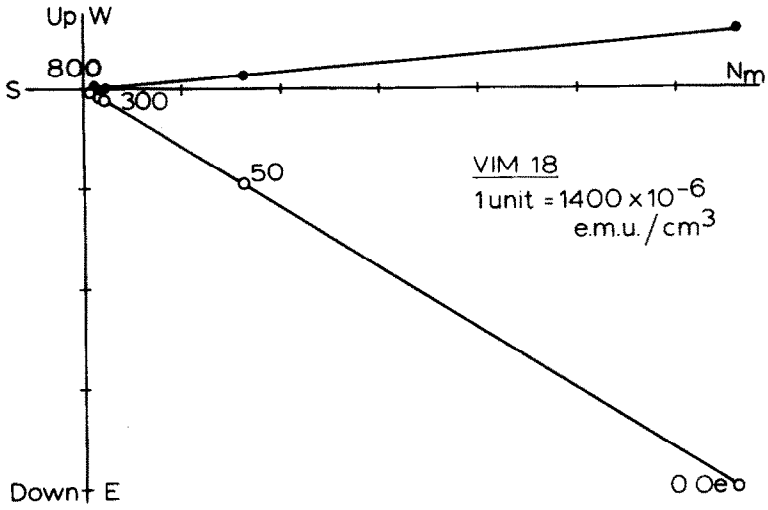


Fig.27. Diagram of a progressive a.c. magnetic field demagnetization of an "unstable" basalt sample from site N (Fig.23). For explanation of the plotted symbols see Fig.3.

strata, being different for each site, thus proves the characteristic magnetizations to be pre-tectonic. (Sites M and O had dips of 15° towards the southeast, site I had a dip of 5° towards the southeast, and sites K and L had dips of 5° towards northeast and east.) It is, therefore, justified to call these magnetizations primary or original.

Localities and geologic situation of the Monchique samples (southern Portugal)

For comparison with the basalts and granite from the Lisbon region, a visit was made to an intrusive syenite body near Monchique (southern Portugal). This syenite complex intruded into Devonian and Carboniferous sediments (McGillavry, 1961). It has recently been dated by Priem et al. (1967) on the basis of the whole rock K-Ar method as 57 m.y. old. According to the synthetic time scale (Échelle synthétique, 1966) and the time scale of the Holmes' Symposium (see International Union of Geological Sciences, 1967) this is a Paleocene–Eocene age.

From two sites eight samples of fresh unweathered rock have been collected. One site was situated 1.5 km south of Monchique on the main road towards Portimao, the other site 5 km south of Monchique on the same road.

Analysis of the N.R.M. of the Eocene syenite of Monchique

The intensities of the total N.R.M. of these samples were about $100 \cdot 10^{-6}$ e.m.u./cm^3. They had Q-values of about 0.2. Apart from the relatively large induced magnetization, all samples had a prevailing "soft" remanent magnetization, concealing the ancient N.R.M. In the demagnetization diagram of Fig.28 it can be observed that the elimination of this

TABLE III

Pole positions of the rocks from Africa and Europe, as far as they have been used for comparison with the results from the Iberian Peninsula

No.	Formation	Age*1	Test*2	α_{95}*3	Pole position	Reference*4
Africa:						
III	Shawa ijolite, Rhodesia	Trm, 209 m.y.	a.c.	11.5	64°N 94.5°W	Briden, 1967 (B 7)
II	Zambia redbeds	Tru	a.c., th.	4.5	68°N 130.5°W	Briden, 1967 (B 8)
I	Ecca redbeds, Songwe, Tanganyika	Pl	th.	12	27°N 91°W	Briden, 1967 (B 6)
Europe:						
44	Georgian volcanics and sediments, U.S.S.R.	Te	-	15 (11)	75°N 123°W	Irving, 1964 (11.117)
43	Turkmenian sediments, U.S.S.R.	Tpa-o	-	-	60°N 158°E	Irving, 1964 (11.044)
42	Turkmenian sediments, U.S.S.R.	To	-	-	70°N 162°E	Irving, 1964 (11.045)
31-41	British Isles, volcanics combined	Te-o	- (a.c.)	4 (11)	78°N 153°E	Irving, 1964 (11.026)
30	Bulgarian andesites	Ku	a.c. th.	-	66°N 177°E	Vollstädt et al., 1968
29	Turkmenian sediments, U.S.S.R.	Ku	rev., unf.	-	61°N 167°E	Irving, 1964 (10.05/6)
28	Siberian traps, combined, U.S.S.R.	Tr	-	-	60°N 133°E	Irving, 1964 (8.20)
27	Triassic sediments, U.S.S.R.	Trl	-	-	51°N 159°E	Irving, 1964 (8.14)
26	Arran sandstone, Scotland	Tr	-	-	54°N 118°E	Irving, 1964 (8.08)
25	Keuper marls, England	Tru	a.c., th.	-	44°N 134°E	Irving, 1964 (8.07)
24	Keuper marls, England	Tru	a.c., th.	12 (9)	43°N 131°E	Irving, 1964 (8.06)
23	Buntsandstein, Germany	Trl	-	13 (5)	55°N 169°E	Irving, 1964 (8.04)
22	Vosge sandstone, France	Trl	-	-	28°N 143°E	Irving, 1964 (8.02)
21	Vosge sandstone, France	Trl	-	-	62°N 167°E	Irving, 1964 (8.01)
20	Nideck volcanics, France	Pl	th.	4 (9)	47°N 169°E	Roche et al., 1962
19	Nahe, Winnweiler volcanics, Germany	Pl	a.c.	-	42°N 163°E	Nijenhuis, 1961
18	Nahe, Grenzlager volcanics, Germany	Pl	a.c.	-	48°N 168°E	Nijenhuis, 1961
17	Lodève upper redbeds, France	Pl	a.c.	-	48°N 164°E	Kruseman, 1962
16	Oslo volcanics, Norway	Pl	a.c.	1 (27)	47°N 157°E	Van Everdingen, 1960
15	Lodeve, lower redbeds, France	Pl	a.c.	-	44.5°N 178°E	Kruseman, 1962
14	Czechoslovakian redbeds	Pl	a.c.	3.5 (6)	40.5°N 164.5°E	Krs, 1968
13	Exeter volcanics, England	Pl	a.c., unf.	6.5 (5)	49.5°N 148.5°E	Zijderveld, 1967b
12	Exeter volcanics, England	Pl	a.c., th	9.5 (7)	47.5°N 156.5°E	Cornwell, 1967*5
11	Czechoslovakian redbeds	Cu	a.c.	8 (6)	38.5°N 162.5°E	Krs, 1968
10	Great Whin Sill, England	Cu, 281 m.y.	a.c.	4 (34)	37°N 169°E	Irving, 1964 (7.15)

9	Dikes and sills, Sweden	Cu, 282 m.y.	a.c., th.	2.5	(19)	35.5°N 174°E	Mulder, in prep.
8	Scotland lavas	Dl	a.c., rev.	19	(5)	1°N 121°E }	Creer and Embleton
7	Scotland redbeds	D	th.	11	(10)	2°S 117°E }	1967
6	Old red sandstone; Röragen, Norway	Dl	th.	15	(8)	19°N 160°E	Storetvedt and Gjellestad, 1966
5	Old red sandstone, Kvamshesten, Norway	Dl	a.c., th.	-		22°N 170°E	Lie, 1967
4	Ringerike sediments, Norway	Su	th.	-	(3)	21°N 159°E	Storetvedt et al. 1968
3	Eycott group, England	O	a.c., th.	11	(7)	14°N 165°E	Nesbitt, 1967
2	Builth volcanics, England	O	a.c., th.	-		15°N 162°E	Nesbitt, 1967
1	Arenig lavas, England	O	a.c.	-		11°N 168°E	Nesbitt, 1967

*[1]Te = Eocene; To = Oligocene; Tpa = Paleocene; K = Cretaceous; Tr = Triassic; P = Permian; C = Carboniferous; D = Devonian; S = Silurian; O = Ordovician; u = upper; m = middle; l = lower.

*[2]a.c. and th. (thermal) describe the cleaning techniques, rev. = reversals, unf. = unfolding test.

*[3]Only given for mean directions computed while giving unit weight to site means.

*[4]Letters and numbers between brackets as given by Irving (1964) and Briden (1967) in their review articles.

*[5]Cornwell, 1967, selected data: only site means corrected for the tectonic dip of the strata have been used.

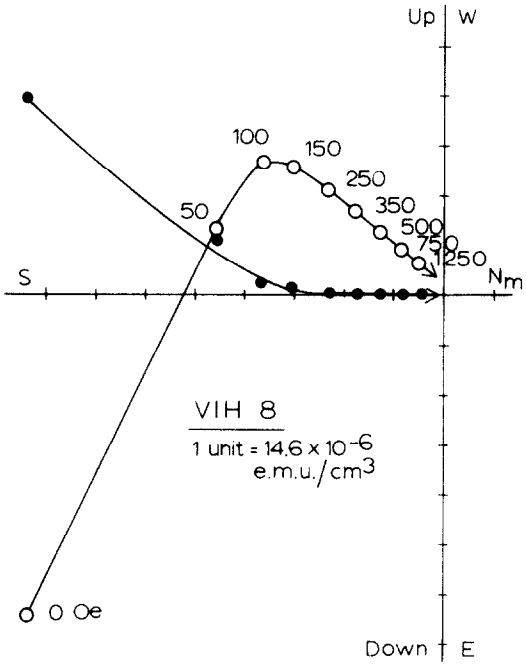


Fig.28. Demagnetization diagram of a syenitic sample from Monchique (southern Portugal). For explanation of the plotted symbols see Fig.3.

Not corrected for the present-day geomagnetic declination (August 1967: 9.2°W).

soft and partly viscous magnetization occurs in the a.c. magnetic field trajectory between 0 and 200 Oe. Between 200 and 1,500 Oe (between 400°C and 580°C in thermal analysis) another component was revealed with a direction similar in all samples, being thus characteristic for the syenite complex. These directions have been plotted in Fig.25B. A mean direction of magnetization has been computed giving unit weight to samples in the analysis, with $D = 182°$, $I = -37°$ and $\alpha_{95} = 6.5°$.

DISCUSSION OF RESULTS

In the section "Localities and analysis of the N.R.M." I have dealt with empirical observations. The following part will be devoted to interpretations made on the basis of these observations. The purpose is to compare the paleomagnetic observations from the Iberian Peninsula (Table II) with those from stable Europe and Africa (Table III). The accuracy of the interpretations obviously will depend on the validity of the data used. It will be necessary, therefore, to review critically all relevant paleomagnetic work in Spain as well as in stable Europe and Africa.

Again I will do this in chronological order, though, unfortunately, in doing so one is dealing first with the more remote and as yet less well known periods of the Paleozoic.

Spain and Portugal

Fig.29 shows in equal-area projection the mean directions of characteristic magnetizations obtained from Ordovician and Silurian rocks of the Iberian Peninsula (triangles). These mean directions have approximately similar inclinations, but show a rather large scatter in declination. This might be due to: (1) secular variation; (2) insecurity about the time of acquisition of the remanent magnetizations; (3) displacements of one part of the Iberian Meseta, relative to other parts, before the end of the Hercynian orogeny.

Each of these factors might have had its influence on my data. *Secular variation* conceivably is of importance in the case of the Ordovician volcanics of Coimbra, since only two samples have been used. So the directions measured may not be representative of the contemporaneous field. *Insecurity of dating* may influence the results of the volcanics of Atienza (Van der Voo, 1967; Table I, no.3), which might be younger than Silurian (see p.11), though the Hercynian orogeny sets an upper limit of Carboniferous. The possibility of *relative internal displacements*, finally, may not be excluded. Before the Hercynian orogenetic cycle there may have been similar movements in the then mobile belts of Europe, as we now are able to trace for the Alpine orogeny in the Tethys zone. For the Early Carboniferous Period this has been suggested by Rutten (1965), and for the Late Carboniferous Birkenmajer et al. (1968) seem to recognize minor rotations in Czechoslovakia.

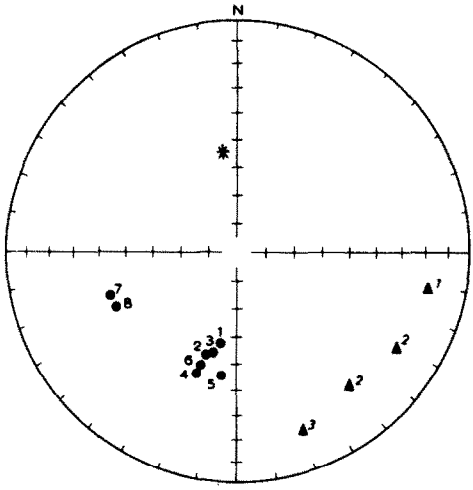


Fig.29. Equal-area projection of the geomagnetic directions for the present-day location of Madrid, computed from the Ordovician, Silurian and Devonian pole positions obtained from European rocks. Triangles: Spain and Portugal, dots and circles: other European countries. The numbers refer to Tables I and II for the Iberian Peninsula and to Table III for the other European countries. Full (open) symbols pointing downwards (upwards), with a positive (negative) inclination. The asterisk represents the present-day local geomagnetic field.

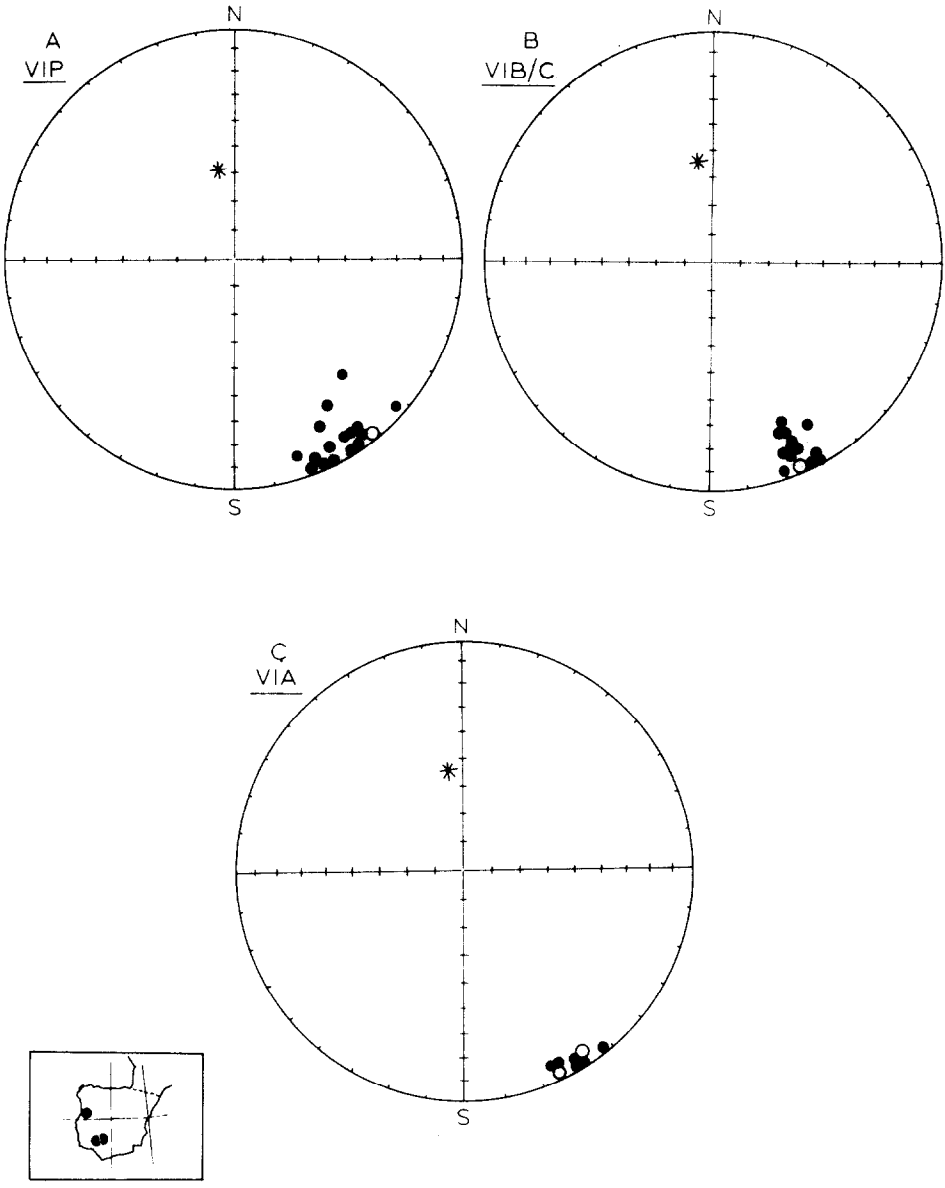


Fig.30. Equal-area projections of the directions of three Upper Carboniferous–Lower Permian formations from the Iberian Meseta, after correction for the tectonic dip of the strata. A. Buçaco redbeds, Table I, no.6; B. Viar dikes and sills, Table I, no.8; C. Viar redbeds, Table I, no.7. For explanation of the symbols see Fig.8.

Other European countries

In the last five years new "cleaning" techniques have caused some uncertainties about the pole positions from Europe of the Earlier Paleozoic, in particular for the Devonian Period (Creer and Embleton, 1967; Storetvedt, 1967). It seems, moreover, to be questionable whether all of the extra-Alpine European block has been a single structural unit before the Hercynian orogeny (Rutten, 1965; Zijderveld, in preparation). For these reasons it is beyond the scope of this paper to draw conclusions from a comparison between northern European data and my own rather scattered results from the Iberian Peninsula. There is no doubt, however, that the results from Spain and Portugal for the Early Paleozoic definitely deviate from those obtained by demagnetization techniques from Belgium (Zijderveld, in preparation), Great Britain (Creer and Embleton, 1967; Nesbitt, 1967) and Norway (Storetvedt, 1967). These data have been summarized in Table III. To obtain data comparable with those from the Iberian Peninsula, the geomagnetic field directions have been computed from these ancient pole positions for the present-day location of Madrid (Fig.29, dots). The western and northern European results all have southwest or south-southwest declinations and positive inclinations, while the data from Spain and Portugal also have positive inclinations, but show southeast or east-southeast declinations.

The Upper Carboniferous and Lower Permian

Spain and Portugal

In the previous section four investigations on Upper Carboniferous–Lower Permian rocks have been described (Table I, no.6–9). Of these the Buçaco Formation of Coimbra is the most important. The age of the magnetization coul be determined as Upper Carboniferous–Lower Permian and since each sample was collected from a different bed it is assumed that the influence of the secular variation has been ruled out.

The sediments and volcanics of the Viar Basin have the same age as the Buçaco Formation. The magnetic directions of these two groups and the Buçaco samples are entirely consistent (Fig.30). This consistency, incidentally, justifies the geologic assumptions of the structural unity of the Spanish Meseta.

In a previous section four investigations on Upper Carboniferous-neous rocks has been carried out by Van Dongen (1967). The mean direction of these Sierra del Cadi volcanics (Fig.31, square) has been summarized in Table II, no.9. This direction deviates by about 15° in declination from the results of the Meseta. This might be caused by relatively small geotectonic movements between Pyrenees and Meseta or by a difference in age. The Sierra del Cadi andesites nonconformably overly Stephanian schists and are overlain by Permo-Triassic redbeds. They might as well be of Middle or even Upper Permian age.

Stable (extra-Alpine) Europe

For the Late Carboniferous–Early Permian Period various reliable observations from Norway, Sweden, England, Germany and northern and southern France are available. In Table III these results have been listed

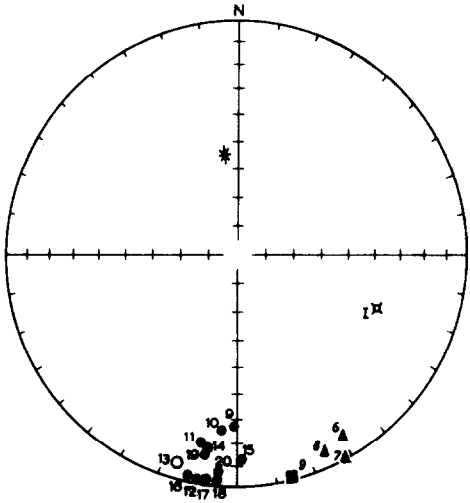


Fig.31. Equal-area projection of the geomagnetic directions for the present-day location of Madrid, computed from Upper Carboniferous–Lower Permian pole positions. Results from: the Iberian Meseta (triangles); the Spanish Pyrenees (square); stable (extra-Alpine) Europe (dots and circles); the African Shield (crosses).

Numbers refer to Tables I, (or II) and III. Full (open) symbols pointing downwards (upwards) with positive (negative) inclinations.

so far as they have been obtained by a.c. or thermal demagnetization techniques.

The pole positions show little scattering as has frequently been remarked by various authors (a.o., Irving, 1964; Krs, 1968). They represent the geomagnetic field at the end of the Hercynian orogenetic cycle and justify the geologic assumption of the structural unity of the Meso-Europe of Stille, also called "stable (extra-Alpine) Europe", since the Late Paleozoic Period. To compare these data with the results of the Iberian Peninsula their age has to be the same, i.e., Stephanian (Upper Carboniferous) or Autunian (Lower Permian). For this reason definitely older or younger results are not included (for instance the Lower Carboniferous Kinghorn lavas, or the Upper Permian Estérel volcanics).

For a comparison, the usual way of plotting the pole positions on part of the globe is less successful in my case. I have chosen two methods of depicting the relationships between results from stable Europe and the Iberian Peninsula: the paleo-isocline map and the equal area projection of the geomagnetic directions. The paleo-isocline map has been constructed from an Early Permian pole position of stable Europe (Fig.32), by computing the location of the paleo-equator and the lines of equal inclination. The direction of the arrows, plotted where the rocks were collected represents the declination, whereas the inclination is also indicated. It is obvious that the Iberian data have deviating (anomalous) declinations, because they are not perpendicular to the paleo-isoclines. Their inclinations,

however, fit well into the paleo-isoclinal pattern. The second method is by computing the geomagnetic directions for the present location of Madrid (3°40'W 40°20'N) from the ancient virtual pole positions. These directions are plotted in one figure (Fig.31). In doing so one can compare the "real" Iberian observations (triangles) with the "virtual" or extrapolated observations from stable Europe (dots and circles). Again one sees that the data from stable Europe and the Iberian Peninsula only deviate in declination for the Upper Carboniferous–Lower Permian. In the following section we will see that this leads to the conclusion that a counterclockwise rotation of the Iberian Peninsula, relative to stable Europe, must have occurred.

The same procedure is followed with the African results (Fig.31, cross). Here there is considerable divergence both in declination and in inclination, which means that not only a relative rotation but also large mutual displacements must have occurred between Africa and the Iberian Peninsula since the Permian Period.

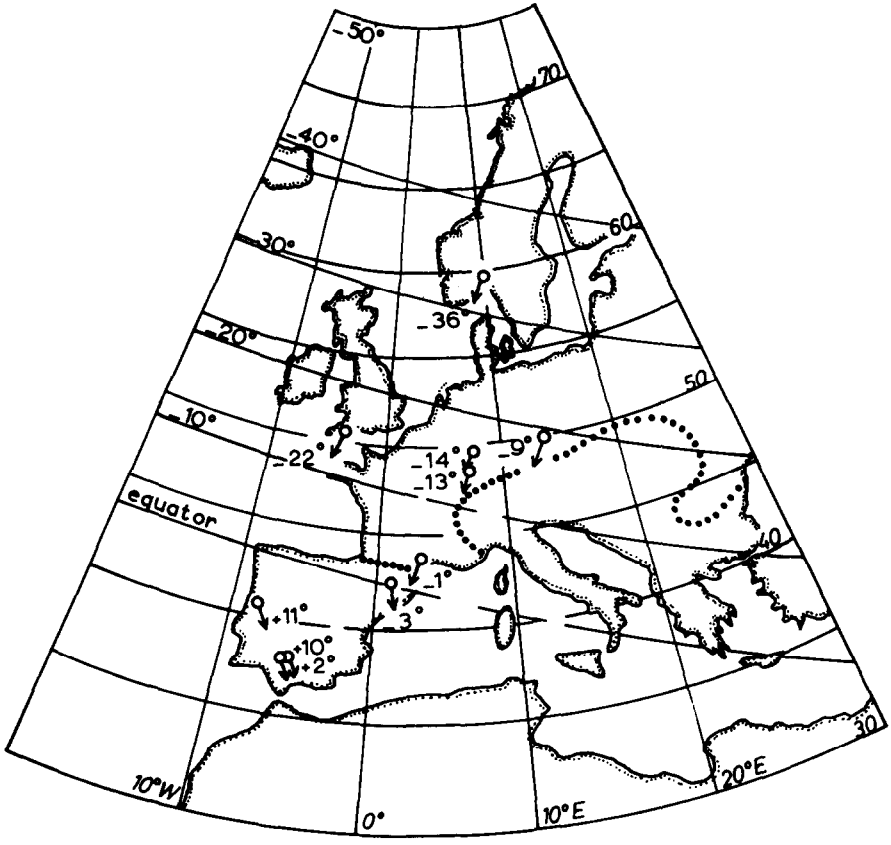


Fig.32. Paleo-isocline map for stable Europe deduced from an Early Permian pole position for stable Europe at 46.5°N, 165.5°E. The direction of the arrows, plotted where the rocks were collected, represents the declination. The inclination is indicated. The dotted line shows the southern boundary of stable (extra-Alpine) Europe.

The Iberian Peninsula

For the Spanish Meseta, the redbeds of Alcázar de San Juan (Table I, no.20) were the only Triassic rocks to yield a characteristic direction of magnetization. Other investigations (Table I, no.14–18) revealed only (secondary) present-day geomagnetic field directions.

In the Spanish Pyrenees Permo–Triassic redbeds have been investigated in three areas (Table I, no.10,12,19). Their age may range between Late Carboniferous and Late Triassic, with a hiatus somewhere in the Permian (H. Visscher, personal communication, 1969). Paleomagnetically, however, there is an indication for a Triassic, perhaps an uppermost Permian age. A long period of reversed polarity, the so-called Kiaman Interval, is assumed to last till at least the end of the Middle Permian (Irving and Parry, 1963; McMahon and Strangway, 1968). In the redbeds of Garralda and the Sierra del Cadi mixed polarities occur.

The volcanic rocks of the Huesca province (Table I, no.11,13) are contemporaneous with the Huesca redbeds as Schwarz (1962) demonstrated.

Apart from the uncertainties about the age relationships, it is unknown to what extent the Alpine orogeny might have influenced the Pyrenean directions. It is, therefore, not surprising to observe some scattering of the Spanish results (Fig.33).

Stable (extra-Alpine) Europe

For the Triassic Period few data from stable Europe are available and they have not been submitted to demagnetization techniques, except a result from tectonically disturbed regions in Czechoslovakia (Kotásek and Krs, 1965) and data obtained by demagnetization of some pilot samples from the Keuper marls of England (Clegg et al., 1954; Creer, 1957).

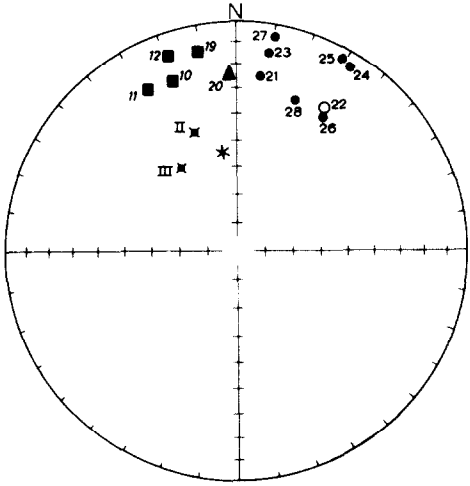


Fig.33. Equal-area projection of the geomagnetic directions for the present-day location of Madrid, computed from Permo-Triassic and Triassic pole positions. For explanation of the symbols see Fig.31.

To compare the results from the Iberian Peninsula, the African Shield and stable Europe, the same procedure is followed as in the case of the Late Carboniferous–Early Permian data. The geomagnetic directions for the present location of Madrid, computed from the ancient pole positions have been plotted in one figure (Fig.33). The geomagnetic directions of the Spanish rocks again display significantly deviating declinations. They have, however, more or less the same inclinations as the directions obtained from the results found for stable Europe.

The Upper Cretaceous and Lower Tertiary

Portugal

The three mean directions of magnetization of the Portuguese rocks (Table I and II, no.21,23,24), no farther apart in age than approximately 25 m.y. are similar (Fig.34, triangles). It has been demonstrated that the magnetization of the Eocene Lisbon Basalts is pre-tectonic. It is, therefore, of Early Tertiary origin. All three directions, furthermore, diverge significantly from the present-day local geomagnetic field direction (Fig.34, asterisk).

Carey (1958) has suggested that the Mesozoic rocks and its basement, west and north of Lisbon, might have been translated relative to the Iberian Meseta. The transcurrent fault zone, along which this movement should have occurred, was called the "Lisbon Scarp". One might, therefore, raise objections to the assumption that the Upper Cretaceous Sintra Granite and the Eocene Lisbon Basalts form part of the Iberian Meseta. The Monchique Syenite, however, intruded into the Carboniferous of the Meseta and the consistency between its mean direction and those of the granite and the basalts does not support Carey's suggestion (Table II, no.21,23,24).

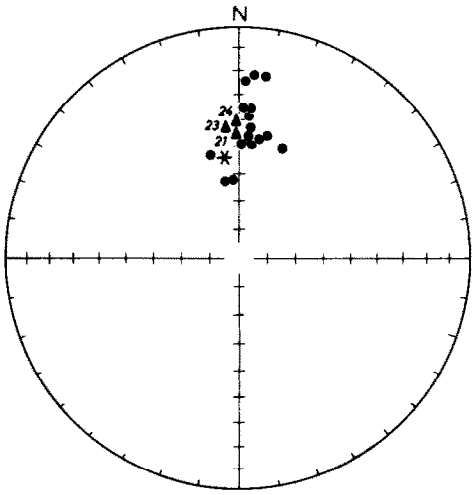


Fig.34. Equal-area projection of the geomagnetic directions for the present-day location of Madrid, computed from Cretaceous and Early Tertiary pole positions. The numbers of the Iberian results (triangles) refer to Table II; the results from stable Europe (dots) are listed in Table III (no.29–44).

Stable (extra-Alpine) Europe

The data for this period obtained from stable Europe are as yet far more inaccurate than those from Portugal. Though for some pole positions unfolding tests indicated reliability, most of the results published have not been obtained by demagnetization techniques.

In the equal area projection of Fig.34 the same procedure has been followed as with the Paleozoic and Triassic results: the dots represent the virtual geomagnetic directions in Madrid, computed from the virtual pole positions for the Cretaceous, Eocene and Eocene–Oligocene of stable Europe, as listed in Table III after Irving (1964).

Because of the above-mentioned uncertainties about the reliability of the results from stable Europe, a comparison with the Portuguese rocks is unsatisfactory. At the moment the only conclusion might be that there is no significant difference between the Late Cretaceous–Early Tertiary Iberian data and the contemporaneous data from stable Europe.

CONCLUSIONS, AND THEIR BEARING ON THEORIES REGARDING CONTINENTAL DRIFT

The well-established Early Permian pole position for stable (extra-Alpine) Europe makes comparisons with the Mediterranean area more successful than for any other geologic period. The Permian paleomagnetism of the Iberian Peninsula is at present the most important result for tectonic considerations.

Combining the results from the Iberian Meseta, I have computed a mean pole position for the Upper Carboniferous–Lower Permian and this implies an ancient geomagnetic field in Madrid with $D = 153°$, $I = +8°$ and $\alpha_{95} = 7°$ ($N = 3$). Extrapolating the paleomagnetic field from the mean virtual pole position for the same period of stable Europe, one would find for Madrid in the position it occupies now a "virtual" paleomagnetic direction with $D = 188°$, $I = +9.5°$ and $\alpha_{95} = 6°$ ($N = 12$). It is obvious that there is no significant difference in inclination, but instead a well-marked difference in declination of 35°. This implies a later counterclockwise rotation of the Iberian Meseta of about 35° with respect to stable Europe. It can be argued that the same implication follows from the Triassic results of Spain (Meseta and Pyrenees), though with far less accuracy (Fig.33).

For the Early and Middle Paleozoic data a comparison encounters large difficulties. Few results from western and northern Europe have been based on demagnetization techniques and the geologic knowledge of the pre-Hercynian geotectonic movements is poor. It is justified to say, however, that marked deviations between Iberian and western European data occur (Fig.29). These deviations point to a counterclockwise rotation of more than 40° for the Iberian sampling areas relative to western Europe since pre-Hercynian time.

The Late Cretaceous and Early Tertiary data from the Iberian Peninsula and from stable Europe do not, on the other hand, display any significant difference. This evidence limits the time for the major part of the postulated rotation of the Iberian Peninsula to the period between Late Triassic and Late Cretaceous. The attribution of an Upper Cretaceous age to sediment in the Bay of Biscay (Cantabria Seamount, Fig.35, black

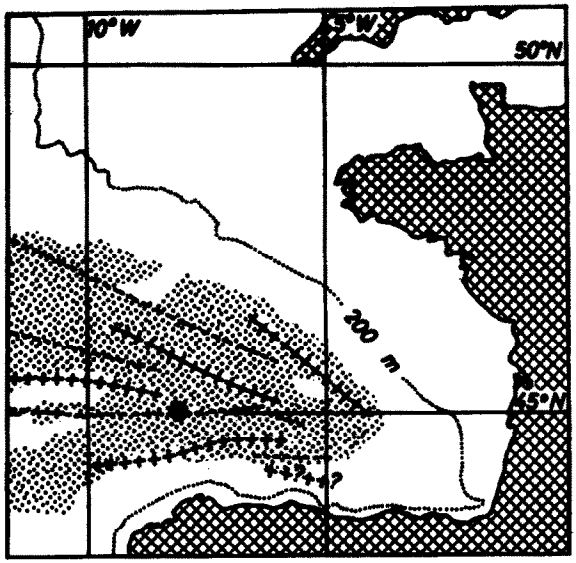


Fig.35. Trends of magnetic anomalies (+ and -) from a survey published by Matthews and Williams (1968). The flat sea floor of the Biscay Abyssal Plain and the lower part of the continental rise (deeper than 4,400 m) is indicated. The black dot is the location of the Cantabria seamount.

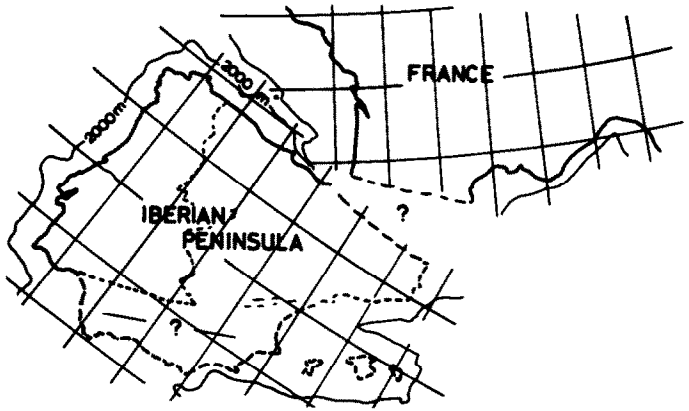


Fig.36. The pre-Triassic position of the Spanish Meseta depicted according to Carey's hypothesis. The future Biscay Sphenochasm still is almost closed at the 2,000 m line, after rotating Spain back over 35°. The present-day contour of the Betic Cordillera (southern Spain) has been dotted since no evidence is available for the former extent or position of this Alpine-folded area.

dot) by Jones and Funnel (1968) supports this dating if we assume that the rotation of the Iberian Peninsula will have been combined with an opening of the Bay of Biscay, as proposed by Carey (1958).

The first to suggest this hypothesis, however, was Argand (1924, p.266, fig.26) who was struck by the tectonic features in France and Spain. He stated: ". . . il faut que la France et l'Espagne, sur l'emplacement actuel du golfe de Gascogne, aient été d'un seul tenant." Carey's study was based on a different approach. He started with a general model by reversing all geologic first order deformations and strains of post-Paleozoic age, thus reproducing a hypothetic paleogeography. In this way *oroclines* (orogenic belts with a change in trend), *sphenochasms* and *rhombochasms* (triangular or parallel-sided gaps in the sialic crust occupied by simatic crust and interpreted as a dilatation) could be defined and hypothetically the tectonic units could be restored into their ancient positions by straightening or, for instance, by rotations. Thereupon a number of applications were tested, among which was the western Mediterranean Basin. For the Iberian Peninsula this resulted in the denomination of the Biscay Sphenochasm and the Betic Cordillera–Riff orocline. Carey proposed a post-Paleozoic rotation of about 35° for the Iberian Meseta around a vertical axis with its pivot point in the western Pyrenees (Fig.36).

This hypothesis by Carey initiated a broad discussion among geologists, in particular those studying the geology of the Pyrenees. Although tectonic movements, for instance along the north Pyrenean fault zone, could be traced for the Mesozoic Period, some authors argued that little evidence could be found for considerable crustal shortening. De Sitter (1965) even defended the hypothesis that the various tectonic lines in northern Spain and the Pyrenees merely separated blocks of the earth crust only differring in their development by vertical movements. The paleomagnetic data, however, are contradictory to these arguments, unless one is willing to place a major tectonic lineament in the poorly known basement of the Aquitain Basin of southwestern France.

More appealing, however, is the theory of a wrench fault character of the north Pyrenean fault zone (Pavoni, 1964; Mattauer, 1968). Translations along this fault would occur parallel to the paleo-isoclines of Fig.32 and would not be revealed by Iberian directions of magnetization. I thus initially favoured a transcurrent movement along a pattern of lineaments: a displacement of the Iberian block along a possible concave zone would cause a counterclockwise rotation. Future work on the paleomagnetism of the rock formations on either side of the north Pyrenean fault zone may eventually provide us with information about the importance of this zone for the rotation of the Iberian Peninsula.

Recently, however, Matthews and Williams (1968) published the results of a magnetic survey in the Bay of Biscay. They report a fan-shaped pattern of linear anomalies above the flat sea floor of the Bay of Biscay, which they relate to the formation of new ocean floor during the rotation of the Iberian Meseta (Fig.35). This evidence highly supports Carey's original hypothesis of a pivot point for the rotation located in the western Pyrenees. So, with the present knowledge Carey's hypothesis is likely to be the most plausible one (Fig.36), although it does not exclude possible transcurrent movements.

I therefore conclude that a counterclockwise rotation of the stable Spanish Meseta of about 35° around a vertical axis, between Late Triassic and Late Cretaceous best explains the paleomagnetic data as related in this study.

REFERENCES

Almela, A. and Febrel, T., 1960. La roca frailesca de Almadén, un episodio tobaceo en una formación basaltica del Siluriano superior. Not. Comm. Inst. Geol. Minero Españ., 59:41–72.

Argand, E., 1924. La tectonique de l'Asie. Proc. Intern. Geol. Congress, 13th, Brussels, pp.171–372.

As, J.A., 1960. Instruments and measuring methods in palaeomagnetic research. Mededel. Verhandel. K.N.M.I., 78:56 pp.

As, J.A. and Zijderveld, J.D.A., 1958. Magnetic cleaning in palaeomagnetic research. Geophys. J., 1:308–319.

Birkenmajer, K., Krs, M. and Nairn, A.E.M., 1968. A paleomagnetic study of Upper Carboniferous rocks from the Inner Sudetic basin and the Bohemian Massif. Geol. Soc. Am. Bull., 79:589–608.

Bonhomme, M., Mendes, F. and Vialette, Y., 1961. Ages absolus par la méthode au strontium des granites de Sintra et de Castro Daire au Portugal. Compt. Rend., 252(21):3305–3306.

Briden, J.C., 1967. Recurrent continental drift of Gondwanaland. Nature, 215:1334–1339.

Bullard, E.C., Everett, J. and Gilbert Smith, A., 1965. A symposium on continental drift. Phil. Trans. Roy. Soc. London, Ser. A, 258:41–51.

Carey, S.W., 1958. A tectonic approach to continental drift. Symp. Continental Drift, Hobart, pp.177–355.

Carrington da Costa, J., 1950. Noticia Sobre Uma Carta Geológica do Buçaco, de Nery Delgado. Direcção Geral Minas Serv. Geol., Lisboa, 28 pp.

Ciry, R., Amiot, M. and Feuillée, P., 1963. Les transgressions Crétacées sur le Massif d'Oroz-Betelu (Navarre Espagnole). Bull. Soc. Géol. France, 7e Sér., 5:701–707.

Clegg, J.A., Almond, M. and Stubbs, P.H.S., 1954. The remanent magnetism of some sedimentary rocks in Britain. Phil. Mag., 45:583–598.

Clegg, J.A., Deutsch, E.R., Everitt, C.W.F. and Stubbs, P.H.S., 1957. Some recent palaeomagnetic measurements made at Imperial College, London. Phil. Mag. Suppl. Advan. Phys., 6(22):219–231.

Cornwell, J.D., 1967. Palaeomagnetism of the Exeter lavas, Devonshire. Geophys. J., 12(2):181–196.

Creer, K.M., 1957. Ac. demagnetization of unstable Triassic Keuper marls from Sw England. Geophys. J., 2:261–275.

Creer, K.M. and Embleton, B.J.J., 1967. Devonian palaeomagnetic pole for Europe and N. America. Nature, 214(5083):42–43.

De Boer, J., 1963. Geology of the Vicentinian Alps (northeastern Italy) with special reference to their palaeomagnetic history. Geol. Ultraiectina, 11:178 pp.

De Boer, J., 1965. Paleomagnetic indications for megatectonic movements in the Tethys. J. Geophys. Res., 70(4):931–944.

De Jong, K.A. and Zijderveld, J.D.A., 1969. Paleomagnetic research in Sardinia, 1. Results of the Alghero section. Nature, in press.

De Sitter, L.U., 1965. Hercynian and Alpine orogenies in northern Spain. Geol. Mijnbouw, 44(11):373–383.

Dietzel, G.F.L., 1960. Geology and Permian paleomagnetism of the Merano region, province of Bolzano, Italy. Geol. Ultraiectina, 4:1–57.

Doetsch, J., 1953. Introducción a un estudio del Término municipal de Pueblo de Guzmán, en la provincia de Huelva. Bol. Inst. Geol. Minero Espan., 65:35–218.

Doubinger, J., 1956. Contribution a l'étude des flores Autuno-stéphaniennes. Mém. Soc. Géol. France, 75: 180 pp.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. Roy. Soc. London, Ser. A, 217: 295–305.

García de Figuerola, L.C., 1959. Localización de un sil en la cuenca del Viar. Estud. Geol., Inst. Invest. Geol. "Lucas Mallada" (Madrid), 15:173–180.

Geologic map of Portugal 1:50,000 (Carta geológica de Portugal). Sheets Sintra, 1960; Cascais, 1935; Loures, 1944; Lisbon, 1950. Serviços Geol. de Portugal, Lisboa.
Geologic map of Spain 1:1,000,000 (Mapa Geológico de España), 1936. 2nd edition, published by Inst. Geol. y Minero España, Madrid.
Grand'Eury, C., 1890. Géologie et paléontologie du bassin houiller du Gard. Théolier, Paris, 354 pp.
Girdler, R.W., 1965. Continental drift and the rotation of Spain. Nature, 207:395–397.
Graham, J.W., 1949. The stability and significance of magnetism in sedimentary rocks. J. Geophys. Res., 54:131–167.
Gregor, C.B. and Zijderveld, J.D.A., 1964. Paleomagnetism and the Alpine tectonics of Eurasia, 1. The magnetism of some Permian red sandstones from north-western Turkey. Tectonophysics, 1(4):289–306.
International Union of Geological Sciences, 1967. A comparative table of recently published geological time scales for the Phanerozoic time - explanatory notice. Geol. Mijnbouw, 46(11):406–409.
Irving, E., 1964. Paleomagnetism and its Application to Geological and Geophysical Problems. Wiley, New York, N.Y., 399 pp.
Irving, E. and Parry, L.G., 1963. The magnetism of some Permian rocks from New South Wales. Geophys. J., 7:395–411.
Jones, E.J.W. and Funnel, B.M., 1968. Association of a seismic reflector and Upper Cretaceous sediment in the Bay of Biscay. Deep-Sea Res., in press.
Klootwijk, C., 1967. Report nr.5 Paleomagnetic Lab. Fort Hoofddijk, State University Utrecht (unpublished).
Koenigsberger, J.G., 1938. Natural residual magnetism of eruptive rocks. Terrest. Magnetism Atmospheric Elec., 43:119–127; 299–320.
Kotásek, J. and Krs, M., 1965. Paleomagnetic study of tectonic rotation in the Carpathian Mountains of Czechoslovakia. Palaeogeography, Palaeoclimatol., Palaeoecol., 1:39–49.
Krs, M., 1968. Rheological aspects in paleomagnetism? Intern. Geol. Congr., 13th, Prague, 1968, Rept., 5:87–96.
Kruseman, G.P., 1962. Etude paléomagnétique et sédimentologique du bassin Permien de Lodève, Herault, France. Geol. Ultraiectina, 9:66 pp.
Lie, L.G., 1967. Paleomagnetiske studier i Kvamshestens Devonfelt. Thesis Univ. Bergen, 95 pp.
Mannot, P.F., 1965. Report nr. 110, Geol. Inst. Utrecht State University (unpublished).
Mattauer, M., 1968. Les traits structuraux essentiels de la chaîne Pyrénéenne. Rev. Géogr. Phys. Géol. Dyn. Ser.2, 10(1):3–11.
Matthews, D.H. and Williams, C.A., 1968. Linear magnetic anomalies in the Bay of Biscay: a qualitative interpretation. Earth Planetary Sci. Letters, in press.
McElhinny, M.W., 1964. Statistical significance of the fold test in paleomagnetism. Geophys. J., 8(3):338–340.
McGillavry, H.J., 1961. The Upper Paleozoic of the Baixo Alentejo, southern Portugal. Congr. Avan. Etudes Stratigraph. Géol. Carbonifère, Compt. Rend., 4, Heerlen, 1958, 395–408.
McMahon, B.E. and Strangway, D.W., 1968. Stratigraphic implications of paleo-magnetic data from Upper Paleozoic–Lower Triassic redbeds in Colorado. Geol. Soc. Am. Bull., 79:417–428.
Mulder, F.G., 1965. Palaeomagnetic investigations in the Vänern district (Sweden). Geol. Mijnbouw, 44:307–312.
Mulder, F.G., in preparation. Paleomagnetic Research in Southern Sweden. Thesis Univ. Utrecht.
Mulder, F.G. and Zijderveld, J.D.A., in preparation. Description of a non-magnetic furnace for thermal demagnetization treatment.
Nesbitt, J.D., 1967. Paleomagnetic evidence for the Ordovician geomagnetic pole position. Nature, 216(5110):49–50.
Nijenhuis, G.H.W., 1961. A palaeomagnetic study of the Permian volcanics in the Nahe region (SW Germany). Geol. Mijnbouw, 40:26–38.

Pavoni, N., 1964. Aktive Horizontalverschiebungszonen der Erdkruste. Bull. Ver. Schweiz. Petrol.-Geol. Ing., 31(80):54–78.
Priem, H.N.A., Hebeda, E.H., Boelrijk, N.A.I.M. and Verschure, R.H., 1967. Report of the ZWO Laboratory for Isotope Geology, Amsterdam. 33 pp.
Roche, A., Saucier, H. and Lacaze, J., 1962. Etude paléomagnétique des roches volcaniques Permiennes de la région Nideck-Donon. Bull. Serv. Carte Géol. Alsace-Lorraine, 15(2):59–68.
Rutten, M.G., 1965. Discussion on "A symposium on continental drift". Phil. Trans. Roy. Soc. London, Ser. A, 258:53–58.
Schroeder, E., 1930. Das Grenzgebiet von Guadarrama und die Hesperischen Ketten (Zentr. Spanien). Abhandl. Akad. Wiss. Göttingen, Math. Physik Kl., Neue Folge, 16:235–291.
Schwarz, E.J., 1962. Geology and paleomagnetism of the valley of the Rio Aragón Subordán north and east of Oza (Huesca province, Spain). Estud. Geol. Inst. Invest. Geol. "Lucas Mallada" (Madrid) 18:193–240.
Schwarz, E.J., 1963. A paleomagnetic investigation of Permo–Triassic redbeds and andesites from the Spanish Pyrenees. J. Geophys. Res., 68(10):3265–3271.
Simon, W., 1940. Variscische Sedimente der Sierra Morena: die Viar Schichten. Senckenbergiana, Lethaea, 26(5):401–426.
Storetvedt, K.M., 1967. A discussion of the Devonian pole for Europe. Tectonophysics, 4(2):155–162.
Storetvedt, K.M. and Gjellestad, G., 1966. Paleomagnetic investigation of an Old Red Sandstone formation of southern Norway. Nature, 212:59–61.
Storetvedt, K.M., Halvorsen, E. and Gjellestad, G., 1968. Thermal analysis of the N.R.M. of some Upper Silurian red sandstones in the Oslo region. Tectonophysics, 5(5):413–426.
Teixeira, C., 1945. O antracolitico continental português. Bull. Soc. Geol. Portugal, 5:139 pp.
Van den Boogaard, M., 1967. Geology of the Pomarao Region (Southern Portugal). Thesis Univ. Amsterdam, 85 pp.
Van der Lingen, G.J., 1960. Geology of the Spanish Pyrenees north of Canfranc, Huesca province. Estud. Geol., Inst. Invest. Geol. "Lucas Mallada", Madrid, 16:205–242.
Van der Voo, R., 1967. The rotation of Spain: paleomagnetic evidence from the Spanish Meseta. Palaeogeography, Palaeoclimatol., Palaeoecol., 3(4):393–416.
Van der Voo, R., 1968a. Paleomagnetism and the Alpine tectonics of Eurasia, 4. Jurassic, Cretaceous and Eocene pole positions from northeastern Turkey. Tectonophysics, 6(3):251–269.
Van der Voo, R., 1968b. Geology and paleomagnetism of lower Triassic sediments in an anticlinal structure east of Atienza (Guadalajara prov., Spain). Geol. Mijnbouw 47(3):186–190.
Van der Voo, R., 1968c. Comments on a paper by N.D. Watkins and A. Richardson on "The paleomagnetism of the Lisbon volcanics". Geophys. J., 16:543–547.
Van Dongen, P.G., 1967. The rotation of Spain: paleomagnetic evidence from the eastern Pyrenees. Paleogeography, Palaeoclimatol., Palaeoecol., 3(4):417–432.
Van Dongen, P.G., Van der Voo, R. and Raven, Th., 1967. Paleomagnetism and the Alpine tectonics of Eurasia, 3. Paleomagnetic research in the central Lebanon Mountains and in the Tartous area (Syria). Tectonophysics, 4(1):35–53.
Van Everdingen, R.O., 1960. Studies on the igneous rock complex of the Oslo region, 17. Paleomagnetic analysis of Permian extrusives in the Oslo region, Norway. Norske Videnskapsakad. I Oslo, Mat.-Naturv. Klasse, 1:80 pp.
Van Hilten, D., 1960. Geology and Permian paleomagnetism of the Val-di-Non area. Geol. Ultraiectina, 5:95 pp.
Van Hilten, D. and Zijderveld, J.D.A., 1966. The magnetism of the Permian porphyries near Lugano (northern Italy–Switzerland). Tectonophysics, 3(5):429–446.
Vollstädt, H., Rother, K. and Nozharov, P., 1968. The paleomagnetic stability and the petrology of some Caenozoic and Cretaceous andesites of Bulgaria. Earth Planetary Sci. Letters, 3:399–408.

Watkins, N.D. and Haggerty, S., 1967. Primary oxidation variation and petrogenesis in a single lava. Beitr. Mineral. Petrog., 15:249–260.
Watkins, N.D. and Richardson, A., 1968. Paleomagnetism of the Lisbon volcanic. Geophys.. J., 15:287–304.
Zbyszewski, G., 1963. Carta geológica dos arredores de Lisboa, folha 4: Lisboa - noticia explicativa. Serv. Geol. Portugal, Lisboa, 35 pp.
Zbyszewski, G., 1964. Carta geológica dos arredores de Lisboa, folha 2: Loures - noticia explicativa. Serv. Geol. Portugal, Lisboa, 38 pp.
Zijderveld, J.D.A., 1967a. A.c. demagnetization of rocks: analysis of results. In: D.W. Collinson, K.M. Creer and S.K. Runcorn (editors), Methods in Paleomagnetism. Elsevier, Amsterdam, pp.254–286.
Zijderveld, J.D.A., 1967b. The N.R.M. of the Exeter volcanic traps (Permian, Europe). Tectonophysics, 4(2):121–154.
Zijderveld, J.D.A., in preparation (a). Paleomagnetism of the Estérel rocks. Geol. Ultraiectina.
Zijderveld, J.D.A., in preparation (b). Paleomagnetism of Devonian rocks from Ardennes, Belgium.
Zijderveld, J.D.A. and De Jong, K.A., 1969. Paleomagnetism of some late Paleozoic and Triassic rocks from the eastern Lombardic Alps, Italy. Geol. Mijnbouw (in press).
Zijderveld, J.D.A., De Jong, K.A. and Van der Voo, R., in preparation. Paleomagnetic research in Sardinia, 2. Permian volcanics of the Lower Coghinas River area (Gallura).